

Crown and bridge

Working Casts and Dies

Lecture 13

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Working Casts and Dies



Working Cast

It is a replica of the prepared tooth (teeth), ridge area and other parts of the dental arch. It is produced from the final impression, on which we will make the wax pattern of the final restoration.



Requirement of Good Working Cast

- 1- Must be free from air bubbles or any distortion.
- 2- Must shows all details and soft tissue reproduction.
- 3- Must have precise articulation.



Requirement of Good Working Cast

- 4- It should be trimmed to ensure adequate access for carving of the wax pattern at the margin and to allow building ceramic or wax from proximal area for perfect contact.
- 5- The remaining unprepared tooth structure cervical to finishing line should be 0.5 – 1 mm visible.



Die

It is the (positive) reproduction of the individual (single) prepared tooth on which we will do wax pattern of a crown or bridge retainers. (It is produced from pouring of final impression).



Types of Dies

Dies are classified according to the materials used for production of the die includes:

- 1- Gypsum.
- 2- Resin.
- 3- Electroplated.
- 4- Flexible.
- 5- Amalgam.
- 6- Metal sprayed die.

1- Gypsum dies

GYPSUM ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) is used widely in dentistry to make dental models.

It is dihydrate form of calcium sulfate.

1- Model plaster (plaster of paris):

Used for pouring primary impressions to produce diagnostic models, as well as the articulating casts.

2- Dental stone:

For use as a working model when a more strong and durable cast is required.



Gypsum dies

3- Die stone, high strength type IV:

It is hemihydrate type with cuboidal shaped particles and reduced surface area produce such properties without thickening of the mix.

4- Die stone, high strength, high expansion type V :

It has higher compressive strength and setting expansion than type IV.

Used for investing purposes.



Advantages of Stone Dies Are

- 1- Easy to be prepared.
- 2- Can be used with all types of impression Materials.
- 3- Cheap.
- 4- Need less equipments and easy to manipulate.

2- Resin dies

EPOXY RESIN:

- Used with rubber base impression materials.
- Available in the form of a paste to which an activator is added to initiate hardening.
- Care should be taken that the activator not come in contact with skin as it is toxic.



Resin dies

Properties:

- 1- Working time 15 min.
- 2- Setting time 1-12 hours according to product.
- 3- Compressive strength after 7 days is 16,000psi.
- 4- Abrasion resistance is superior to stone dies.
- 5- Dimensional change due to polymerization shrinkage is 0.03% -0.3 up for 3 days.
- 6- They are very viscous when poured hence porosity may occur.

Resin dies

Advantages:

- 1- More resistant to abrasion.
- 2- Dimensionally stable.

Disadvantages:

- 1- It cannot be readily introduced into the details of large impression and requires centrifugal machine.
- 2- It shrinks by about 0.1 to 0.2% in about 24 hours.
- 3- It is expensive.

3- Electroplated dies

They are produced when an impression is electroplated by process known as *electroforming*.

Impression compound and elastomeric materials are impression materials which can be electroplated.

Types of dies:

- 1- copper plated dies. It began in the early 1930
- 2- silver plated dies.

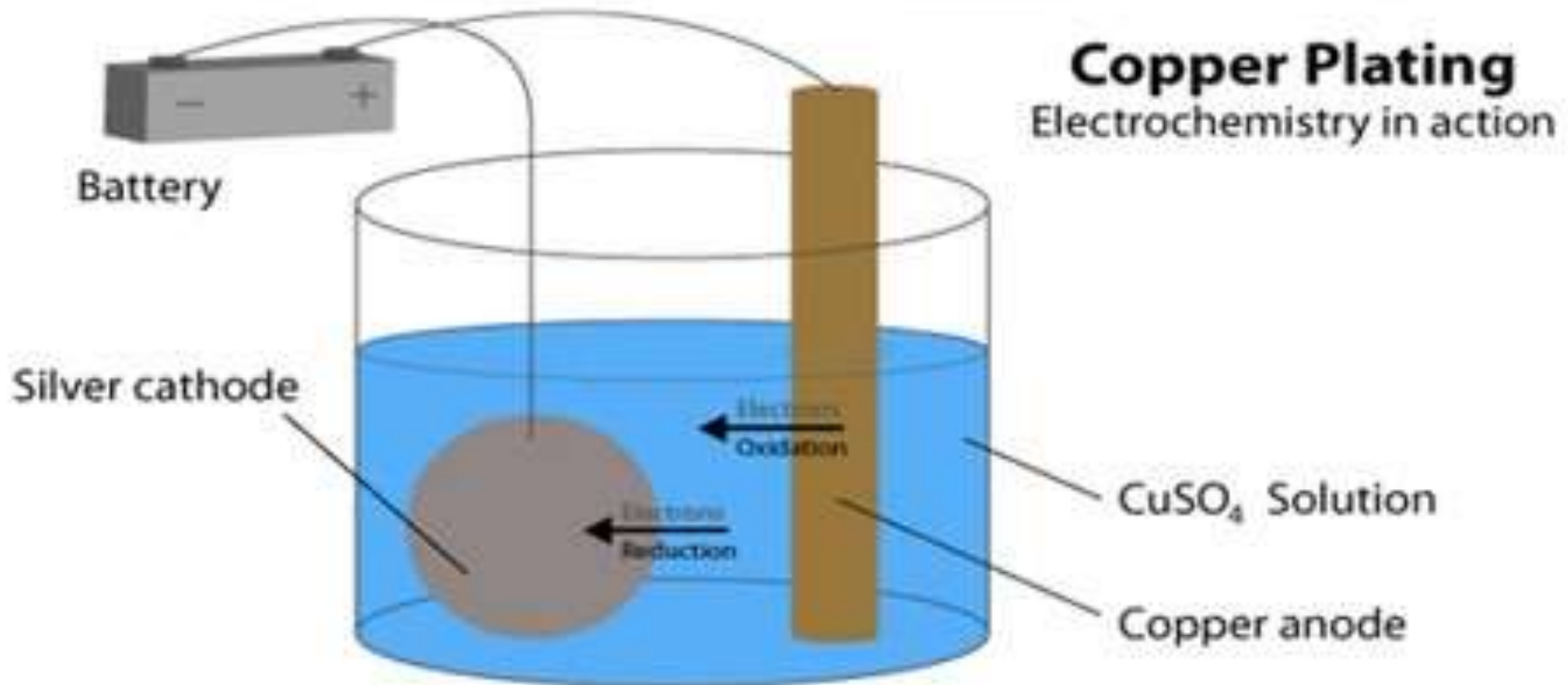
Advantages:

- 1- No dimensional changes occur during electro-deposition of a metal.
- 2- It reproduces the impression accurately.
- 3- The die is tough and has good strength characteristics.

Technique:

- The surface of the impression is coated with a electricity conductor such as graphite, copper powder or silver before it attached to cathode lead wire (*metalizing*). This process determines the surface character of the finished die.
- Copper sulphate is the source of copper, the sulphuric acid increases the conductivity and phenol sulfonic acid helps to assist the penetration of the copper crystals in to the deeper parts of the impression.
- Initially 15 ma current is given which can later be increased to 2 to 3 times the initial current.
- The plating is allowed to proceed for 12 to 15 hours usually (overnight)

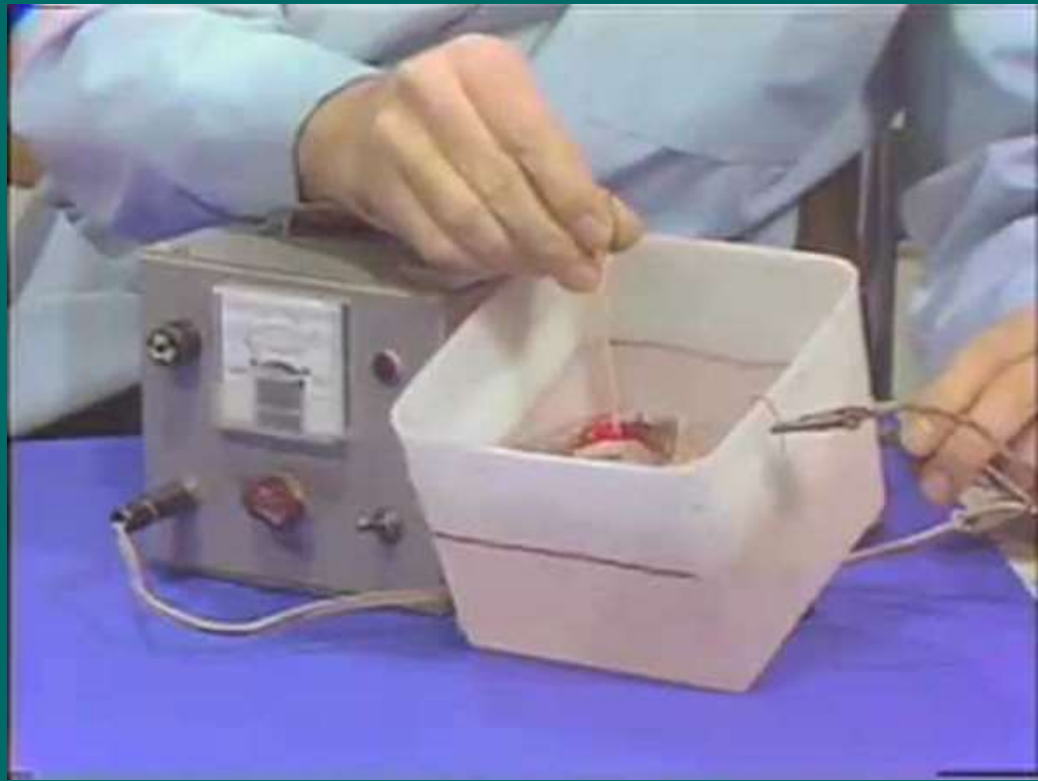
Electroplating



Plated dies:

1- Silver plated.

2- copper plated.



Silver-plated dies

It is done over rubber based impression materials.

It show more vertical change than stone dies with little horizontal changes.

Solution used composed from:

- 1-silver cyanide 36gm.**
- 2- potassium cyanide 60gm.**
- 3-Potassium carbonate 45gm.**
- 4-Water (distilled) 1000ml.**



4- Amalgam dies

- Model amalgam is similar to silver amalgam which is used for filling.
- It is used to make hard metal dies which reproduce the details and sharp margins from impression compound of prepared teeth.
- They cannot be made in hydrocolloid impressions as they cannot withstand pressure.
- after packing the impression with amalgam the die can be removed after a minimum of 12 hours by gently warming the compound.



4- Amalgam dies

- As amalgam is a good conductor of heat, softened wax applied to them cools rapidly.
- This may produce internal stresses which may distort the wax pattern after removal from the die.
- Sudden cooling of the wax may also result in contraction of the wax away from the die.
- A separating agent is needed like with the stone die.



5- Metal sprayed dies

- A bismuth-tin alloy which melts at 138°C can be sprayed directly on to an impression to form a metal shell which can then be filled with dental stone.
- A metal coated die can be obtained rapidly from elastomeric impression material.
- Disadvantage the alloy is soft and care is needed to prevent abrasion of the die.

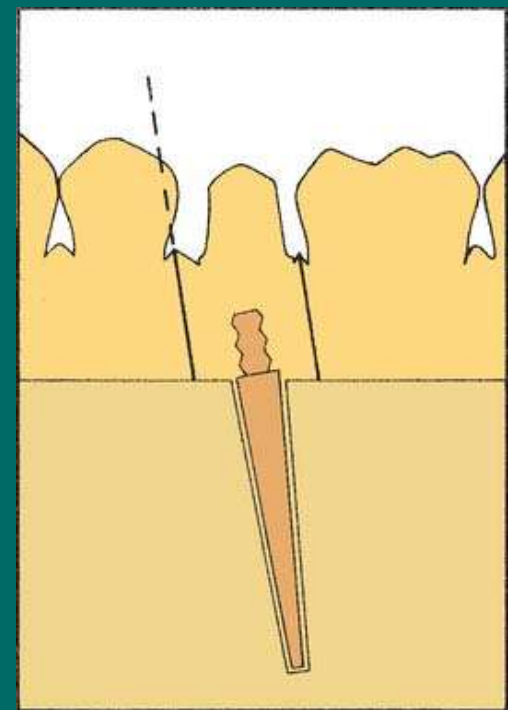
Dowel pin

Ready made metal pins used as a mean for orienting dies to the original model, which allow them to be easily removed and accurately replaced into the model. The pin is tapered and cylindrical but with one flat side for positive seating.



Dowel pin

The brass dowel pin is one of the most accurate dowel types in terms of resisting horizontal displacement and the second lowest in vertical deviation of four types of removable dies



Different types of dowel pins

Dowel pin with projection for simple fixation inside the impression



Different types of dowel pins

Socket type with easy application, no distortion of stone socket and perfect position without rotation



Different types of dowel pins

Metallic pin and socket : no possibility of distortion



Cast and die systems

1) Working cast with removable die:

- I. straight dowel pin.
- II. Curved dowel pin.
- III. Di-lok tray.
- IV. Pindex system.

2) Working cast with separate die.

3) DVA model system.

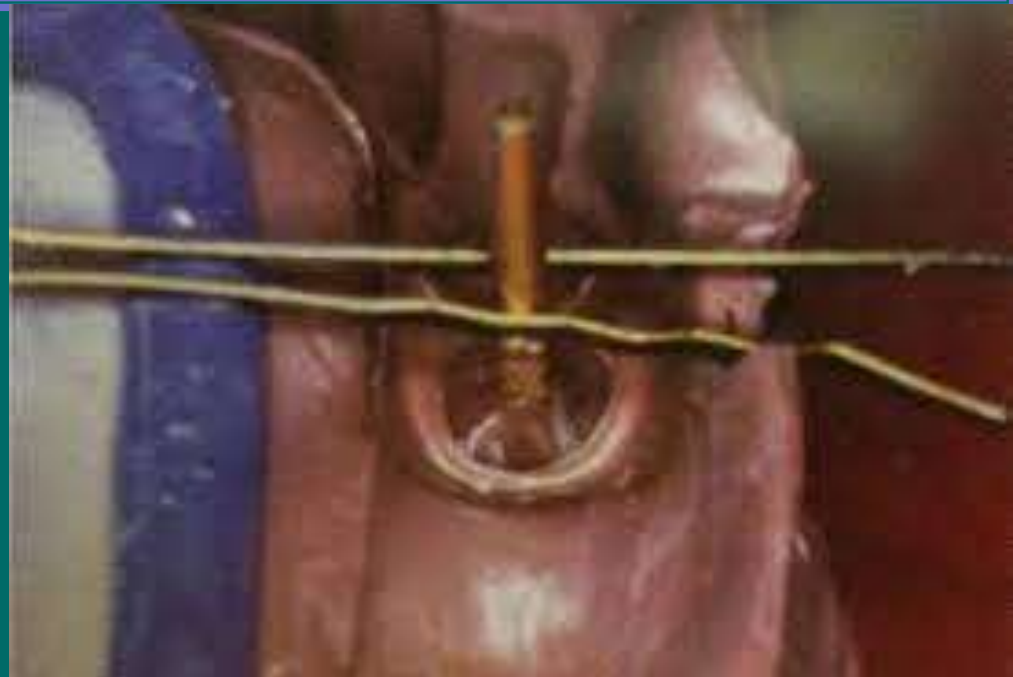
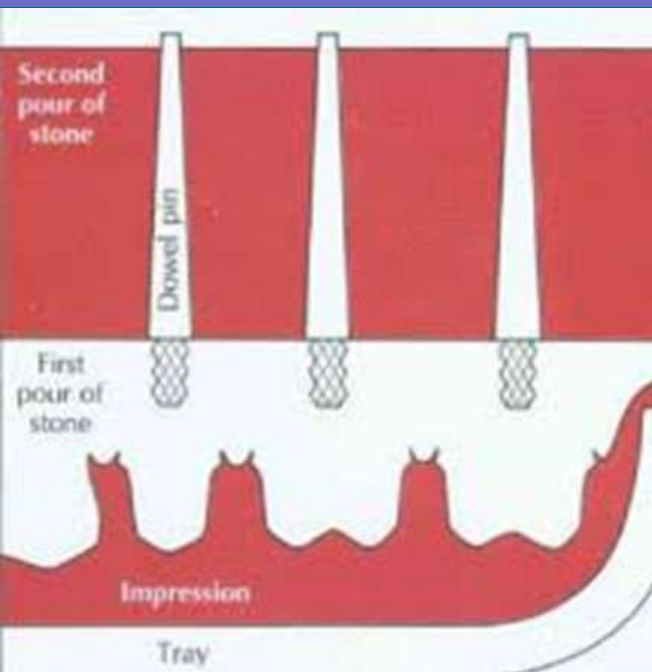
4) Zeiser model system.

Working Cast With Removable Die

1- Straight dowel
pin system

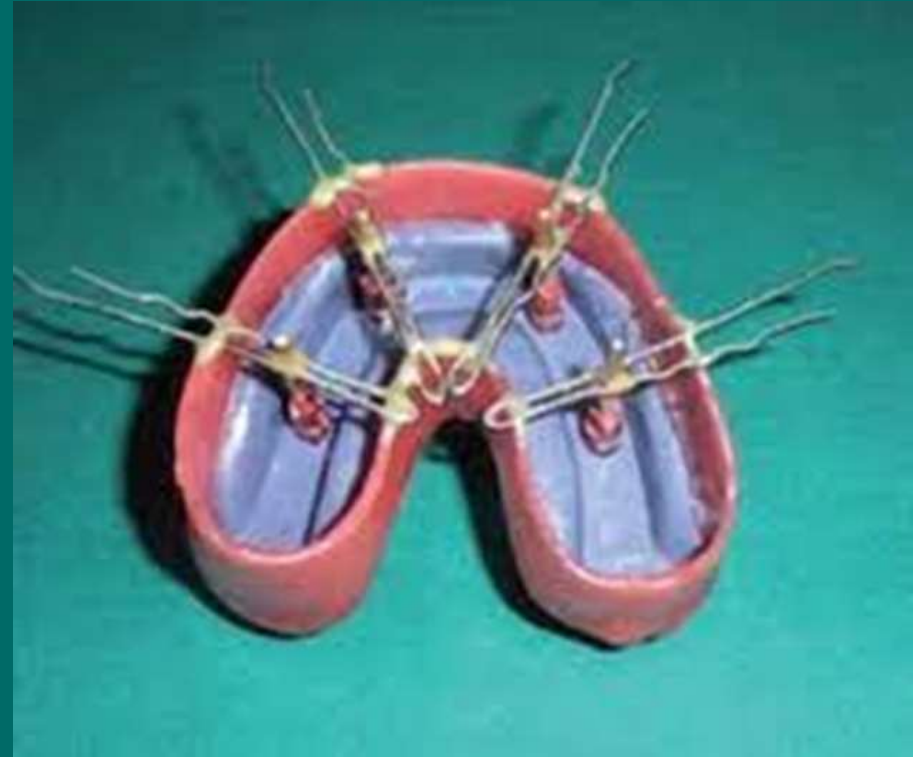
Straight dowel pin system

- Place a dowel pin between the arms of a bobby (hair clip).
- A dowel pin is positioned over each prepared tooth in the impression over the center of the prepared tooth parallel to its long axis.



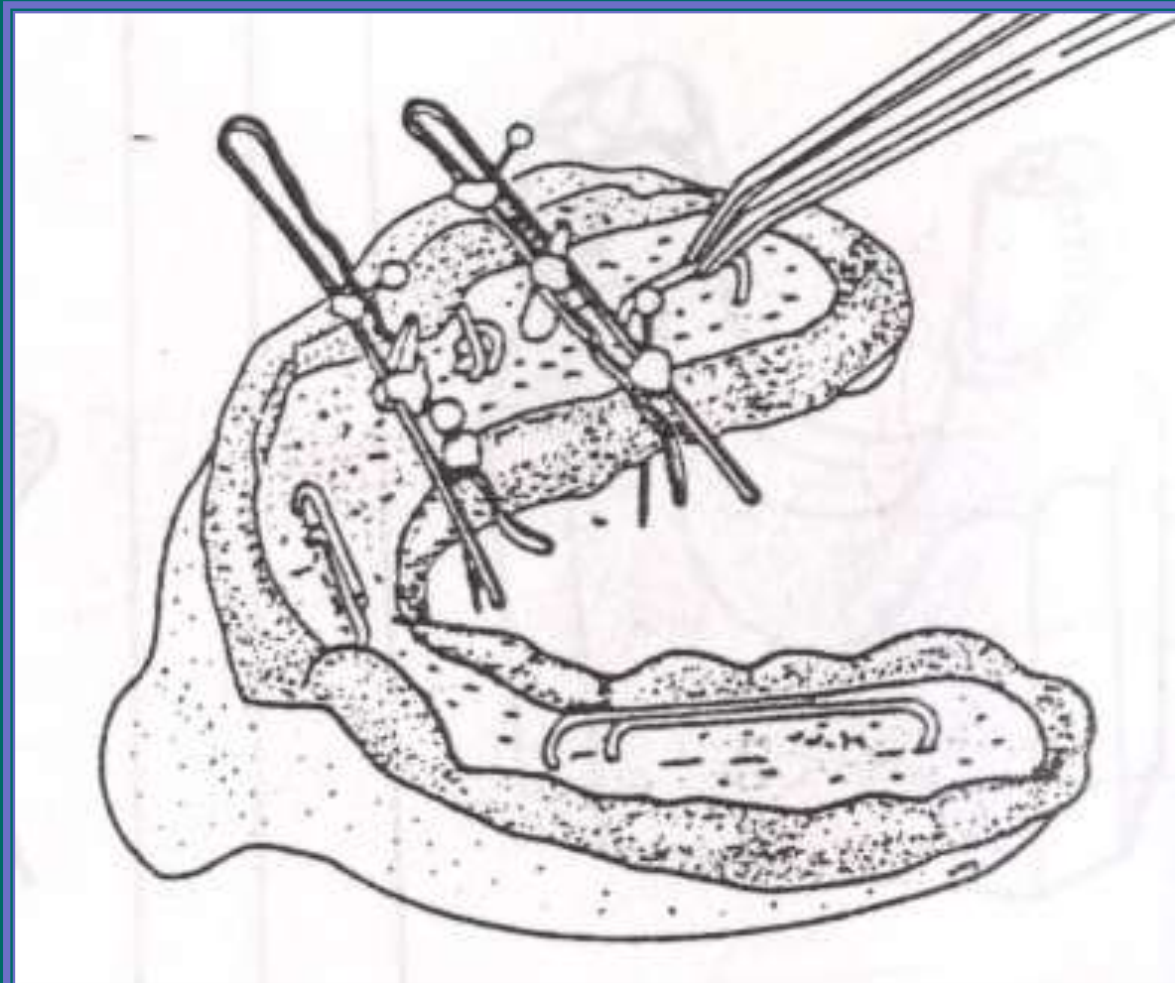
Working Cast With Removable Die

- Pour die stone into the impression, filling the impressions of the teeth and covering knurled (rough) end of the dowel pin.
- The pin should be parallel to the long axis of the preparation.
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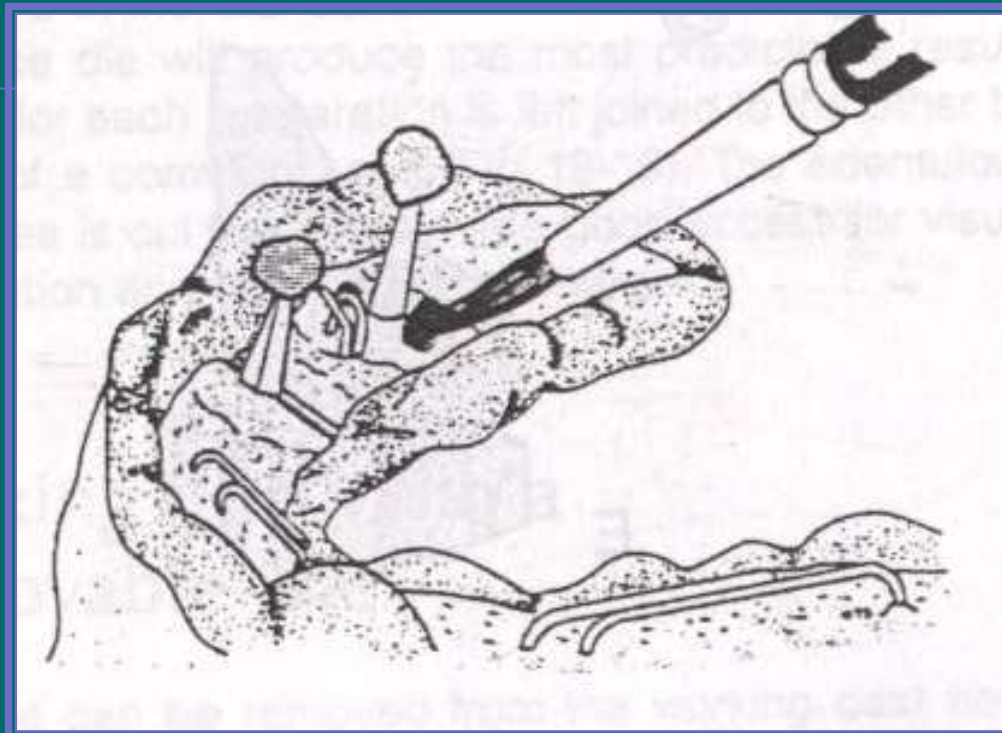
Pouring of working Cast With Removable Die

- A retentive mean is placed (paper clips) in the stone (at the unused areas) before it set.



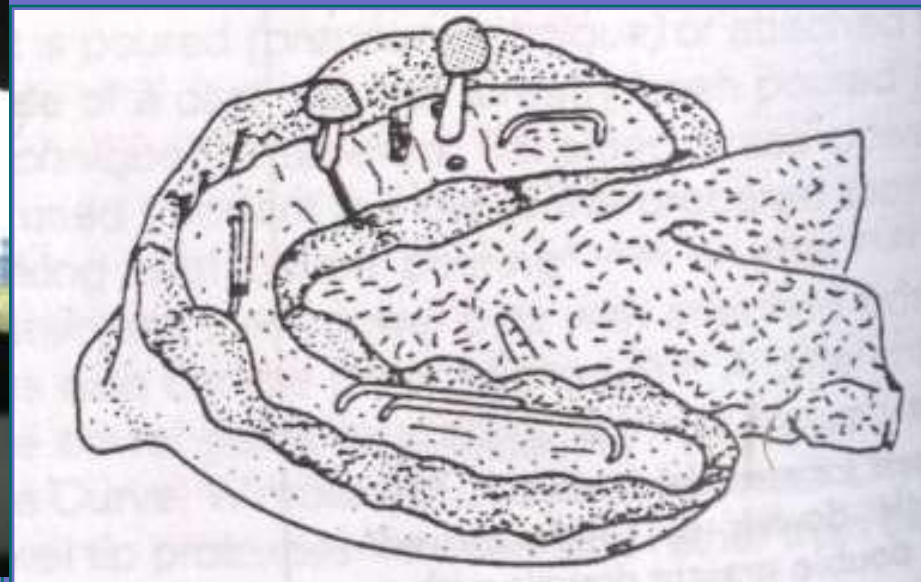
Working Cast With Removable Die

- When stone is set remove the bobby pins.
- Small balls of soft wax is placed on the tip of each dowel.
- Lubricate the surface of the first layer with separating medium around the dowel pin in the area of prepared teeth only.



Working Cast With Removable Die

Pour the 2nd layer (base) with stone till the dowel is completely covered with stone.



Trimming of cast:

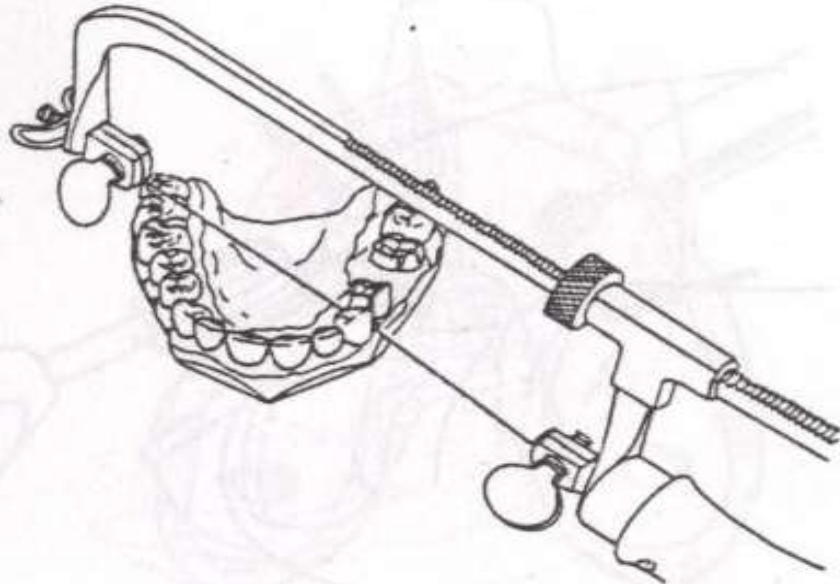
- When the stone sets hard and dry.
- take the cast out of impression.
- Trim the stone cast using electrical trimmer till the wax over the end of dowel pin is shown.



Sectioning of the Cast

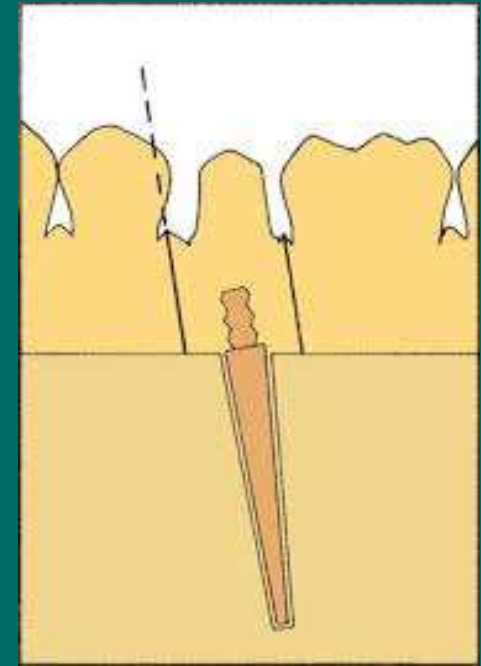
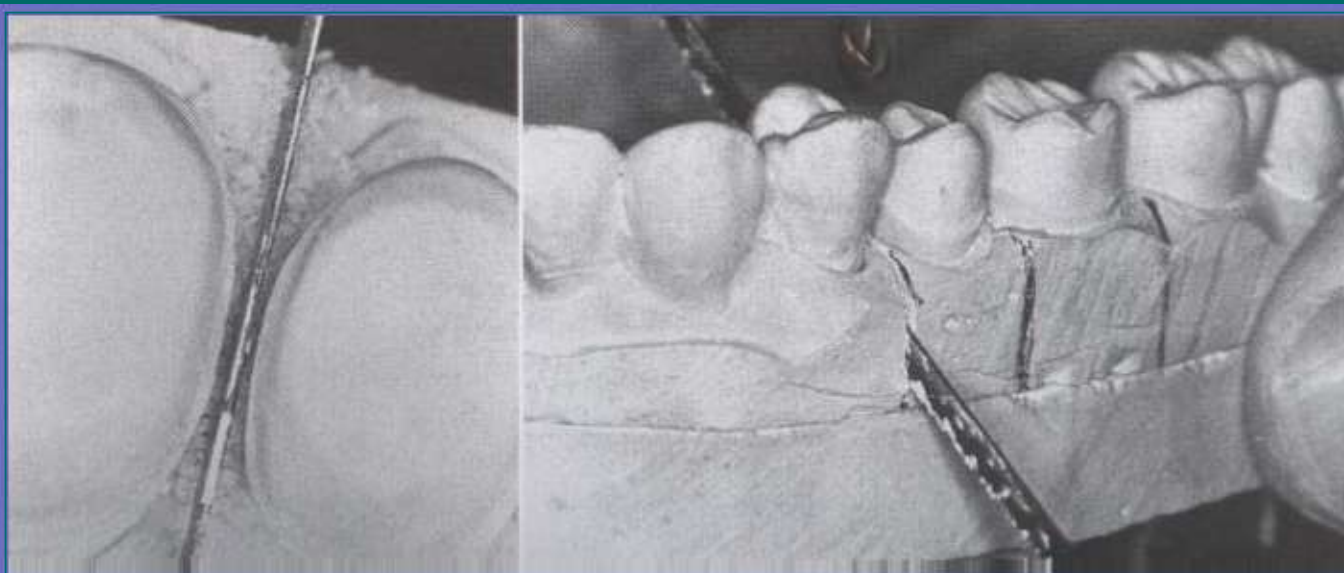
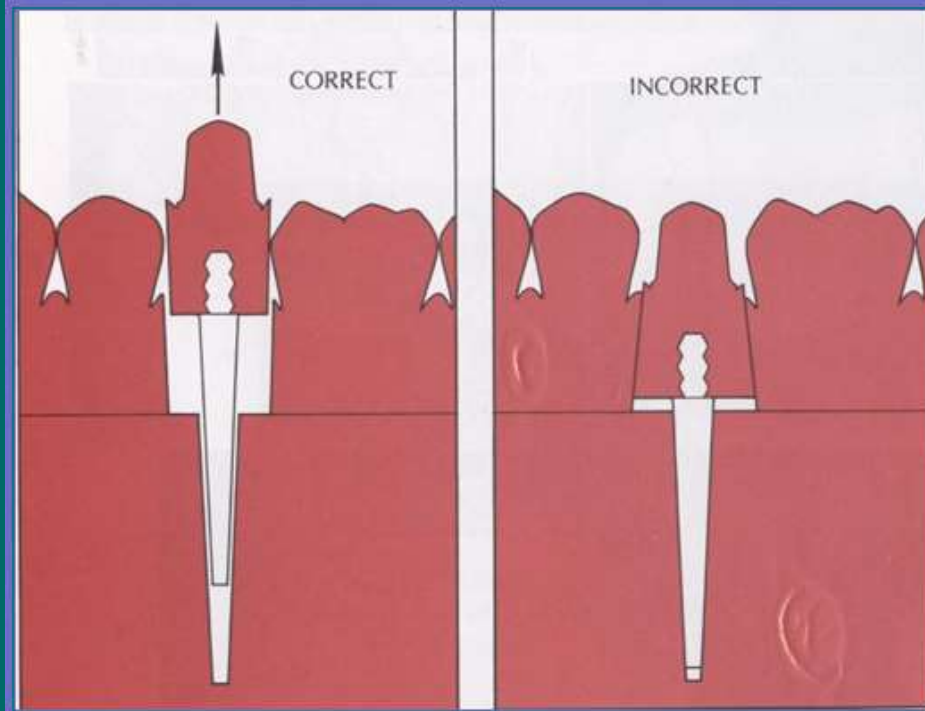


- use a saw frame with a thin blade to separate the die from the working cast .
- with wax knife push the waxed end of dowel pin to get the die out.



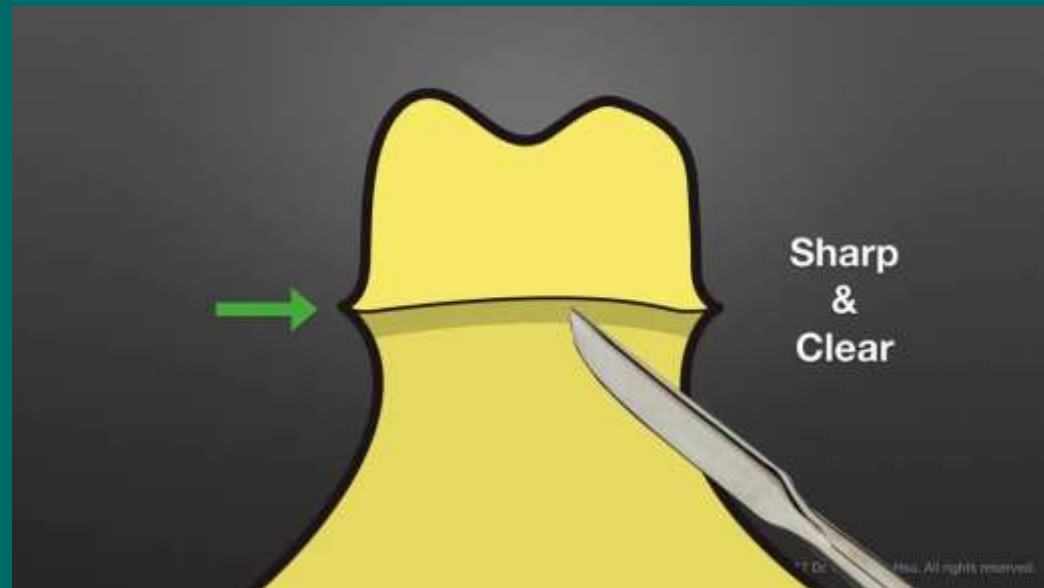
Sectioning of the Cast

- There should be a cut on the mesial and distal side of each die, and the cuts should be vertical and parallel to each other from occlusal to gingivally.



DITCHING OR UNDERCUTTING

- It is trimming of the stone representing the gingiva around the prepared tooth in order to show the finishing line clearly.
- It is done by using a sharp hand instrument or suitable bur.



Preparation of die

- Take the die from the cast and trim away excess stone gingival to the finishing line.
- Complete the trimming of the die and then mark the finishing line with red pencil.
- Repeat the procedure for each die on the cast.



Preparation of die

- Reseat the die on the working cast, place wax ball around the tips of the dowel.
- Soak the cast in water and mount it on articulator using mounting plaster.
- After hardening of plaster remove wax covering the tips of the dowel.

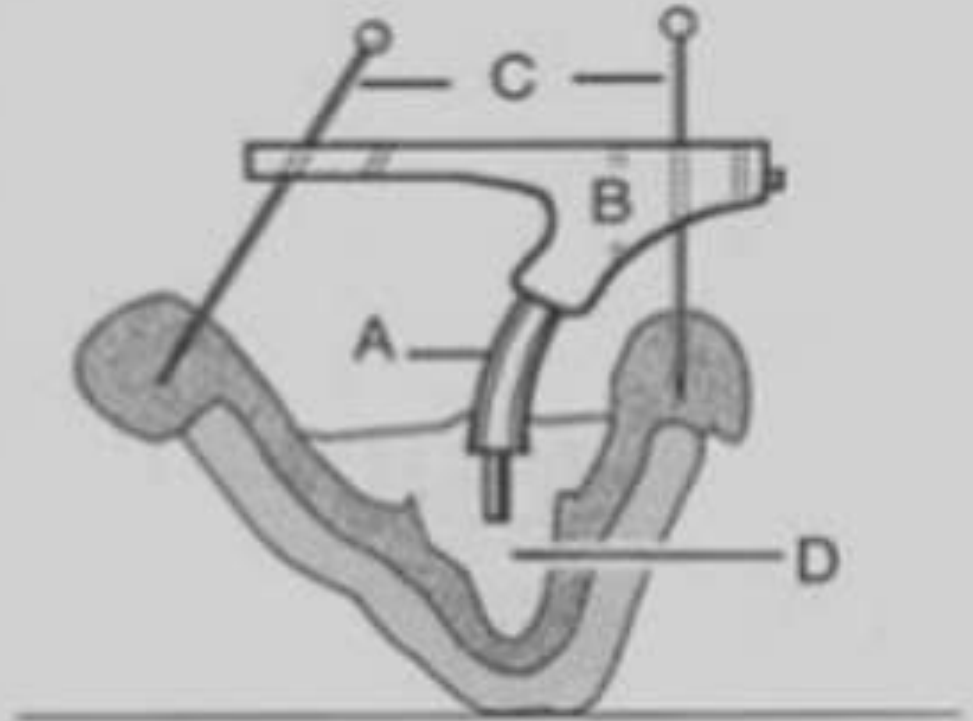


Working Cast With Removable Die

2- Curved dowel pin

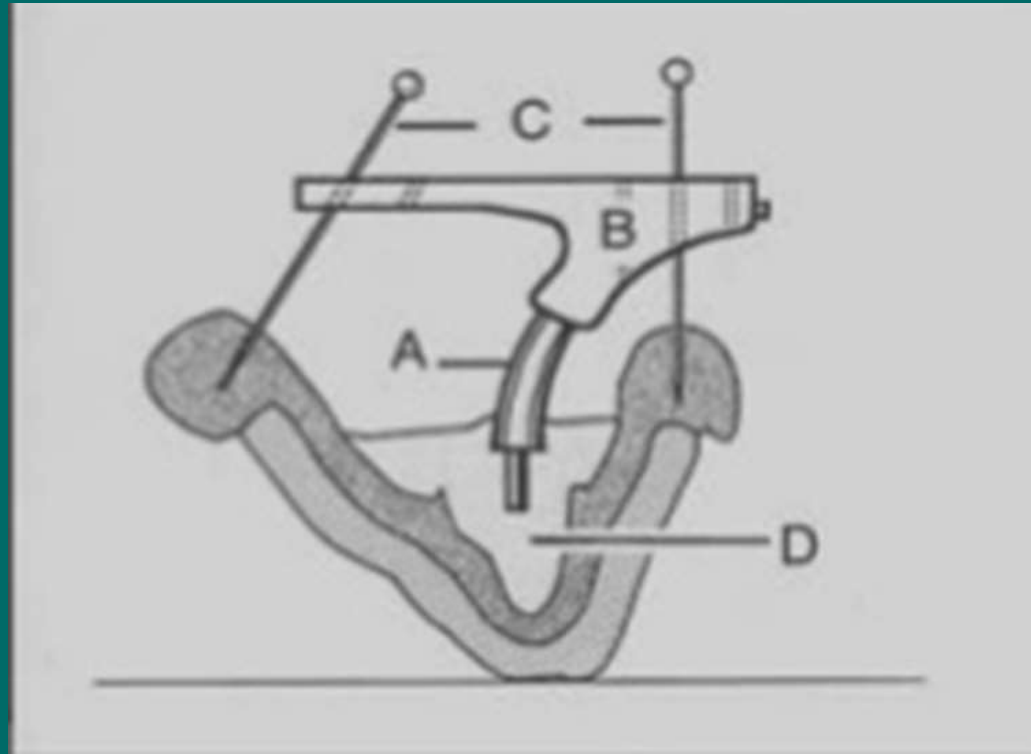
Curved dowel pin

- To install pin before pouring the impression, use finger pressure to insert tip of dowel into large opening of position bar.
- Hold the bar faciolingually, so that the head of the dowel is 1-2 mm into the proposed area of impression.
- The tail of dowel extends facially; if the tooth is linguoversion, turn it towards lingual side for easy removal.



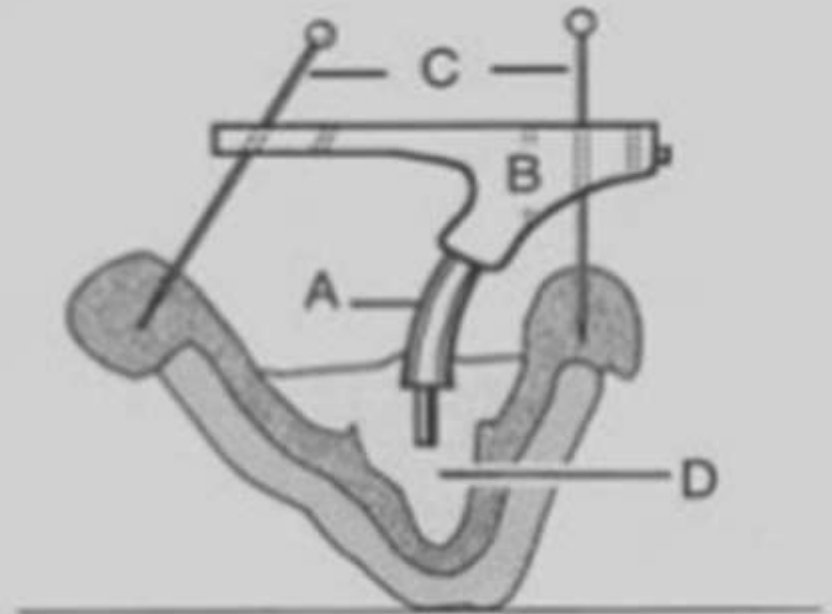
Curved dowel pin

- Insert straight pin (C) into one of 3 holes into the facial aspect of bar and into the facial flange of impression and another pin into lingual flange through lingual holes of bar.
- the dowel should not touch the impression and its head should be parallel to long axis of prepared tooth or teeth.
- This procedure is repeated for all abutments and pontic areas.



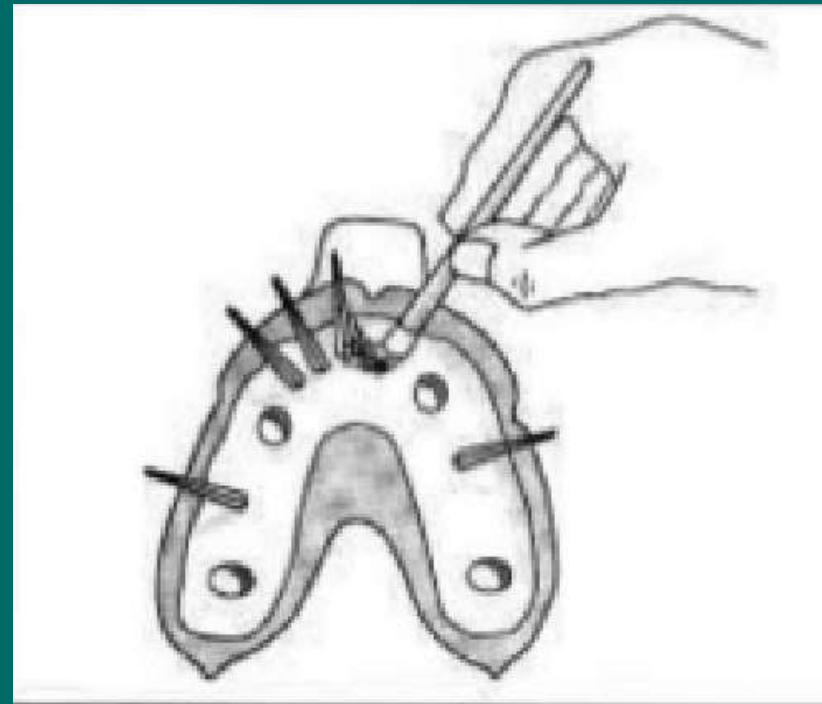
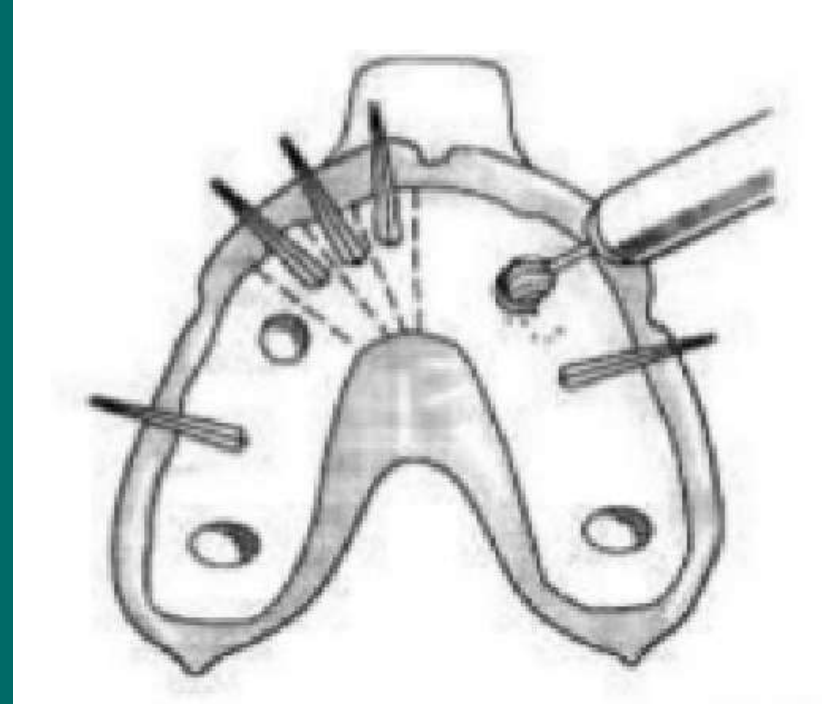
Curved dowel pin

- One pin is placed near the center of each segment of unprepared teeth, this will help in removal of segment for better access of gingival part of wax pattern.
- Impression is poured with die stone until it covers heads of dowel. This will fill the impression about 4 mm above gingival finish line.
- The straight pins and positioning bar are removed as soon as stone is set.



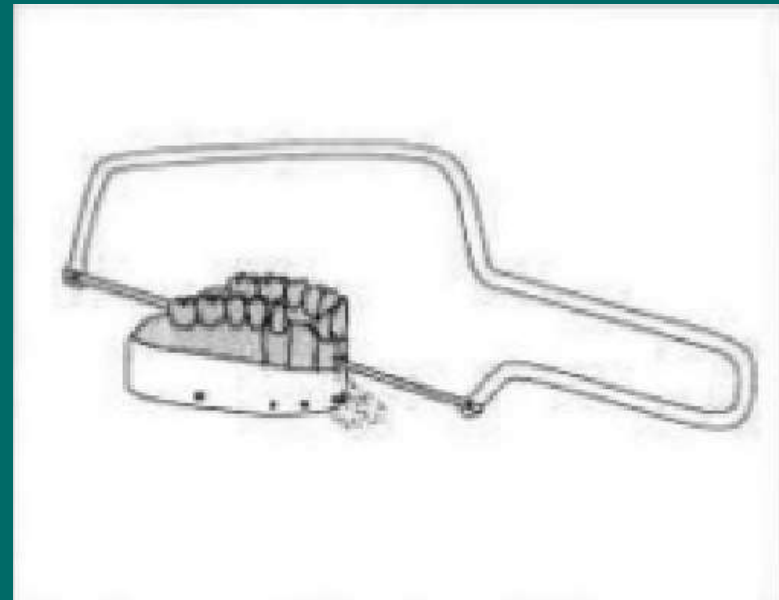
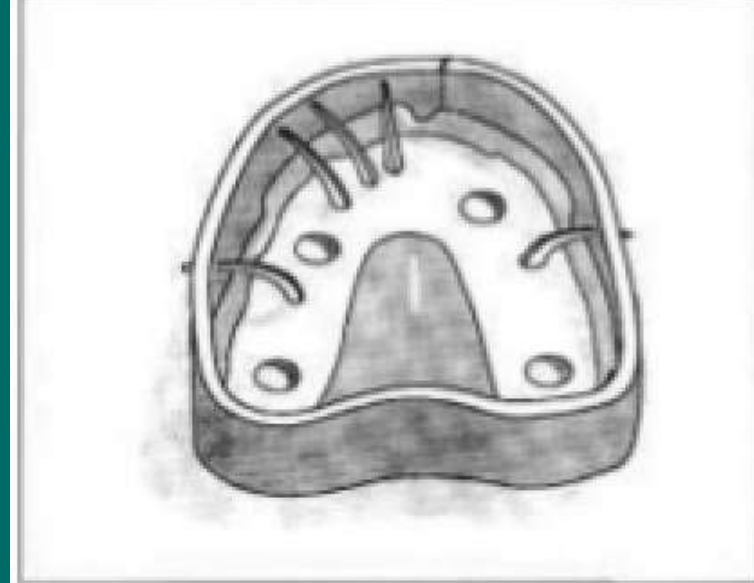
Curved dowel pin

- To assist in orienting each large segments of unprepared teeth . Cut a 2mm deep hole on either sides of each dowel with a large acrylic bur.
- Separating medium is applied on stone and also on exposed part of dowel for easy separation from the base.



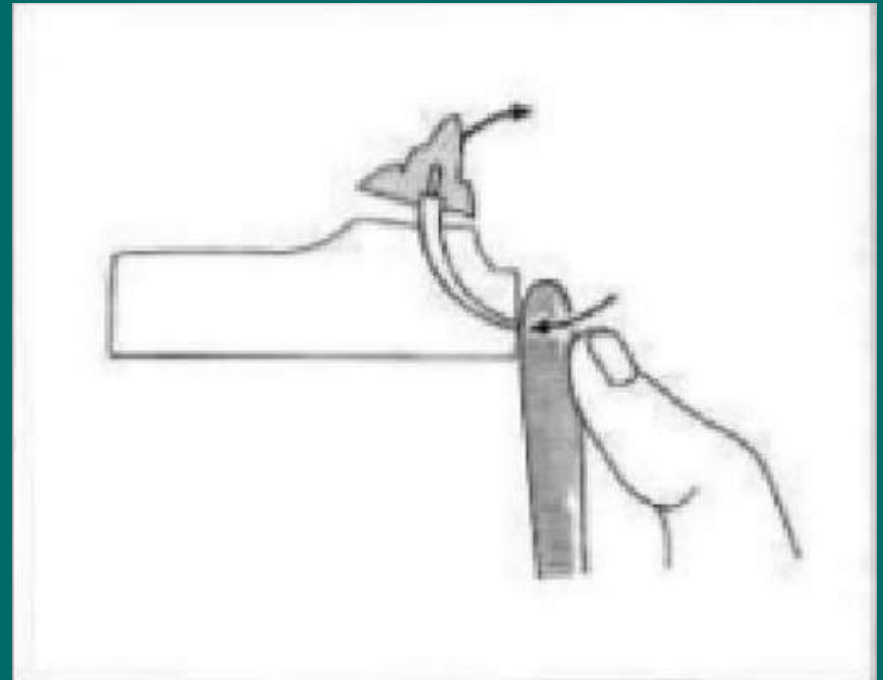
Curved dowel pin

- Impression is boxed allowing the tails of the dowels to extend slightly through the heat softened wax.
- Fill the boxed impression with dental stone. The dowel should be covered by at least 2mm of stone.



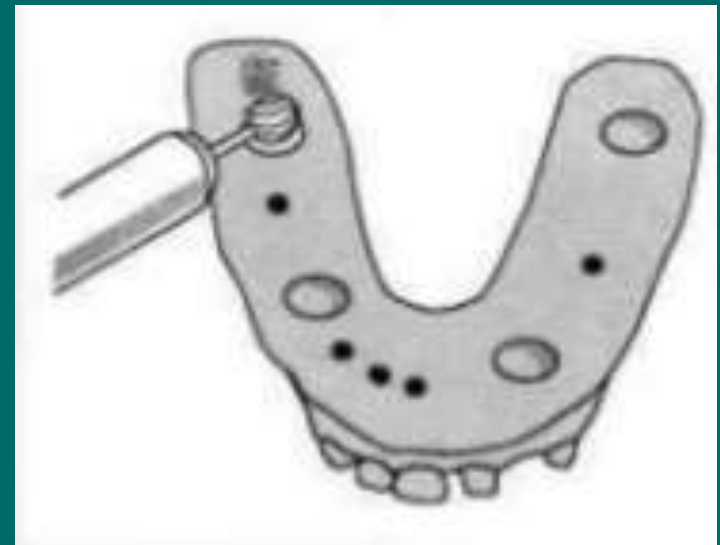
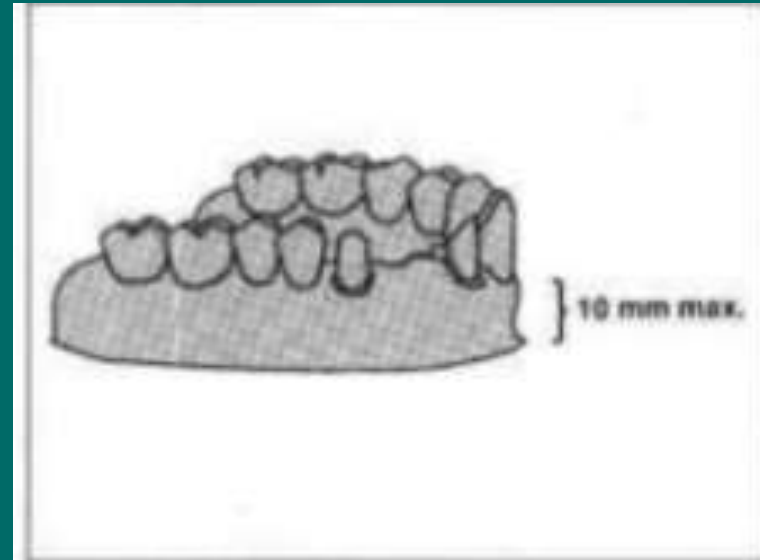
Curved dowel pin

- After the stone has hardened make vertical saw cuts on either side of each die.
- A die is removed by pressing the dowel.



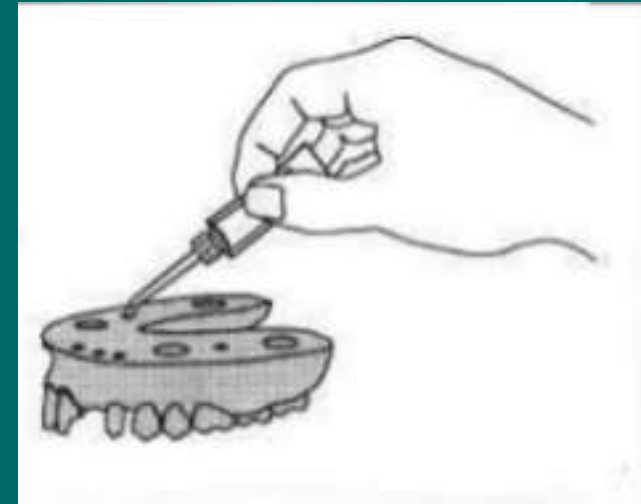
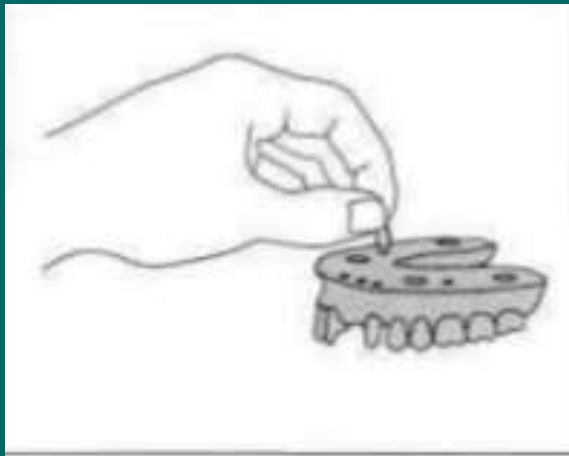
Another method to place curved dowels after the cast has been made

- Pour the impression with die stone to form a horseshoe shaped working cast.
- Trim the bottom of the cast flat to level 10mm from the necks of teeth. This is done because thin cast is easy to saw and short dowels are more stable than longer ones.
- Drill a 5mm deep hole in the bottom of the cast directly under the center of each prepared tooth.
- In areas containing pontics and unprepared teeth can be made with 2mm diameter drill by hand piece.



Another method to place curved dowels after the cast has been made

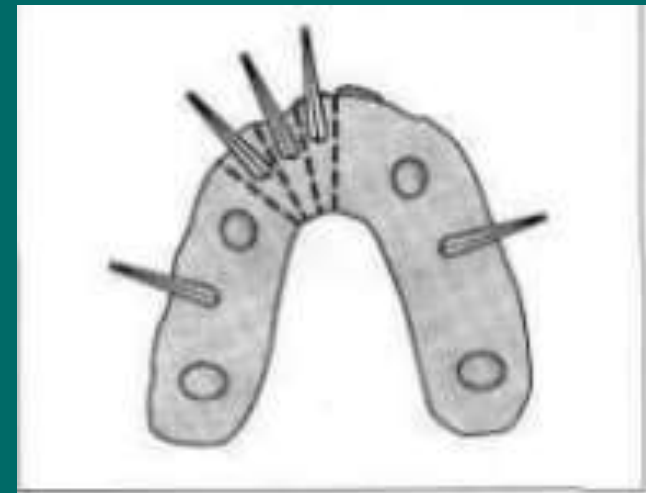
- clean dowel holes then insert curved dowel pin properly and then adjust for proper fit.



Cement dowels into holes by placing a drop of cyanoacrylate cement in each hole.

The head of a curved dowel is seated into the holes. Seat the heads completely with the tail pointing facially.

- Repeat same procedure of pouring base and preparing dies.

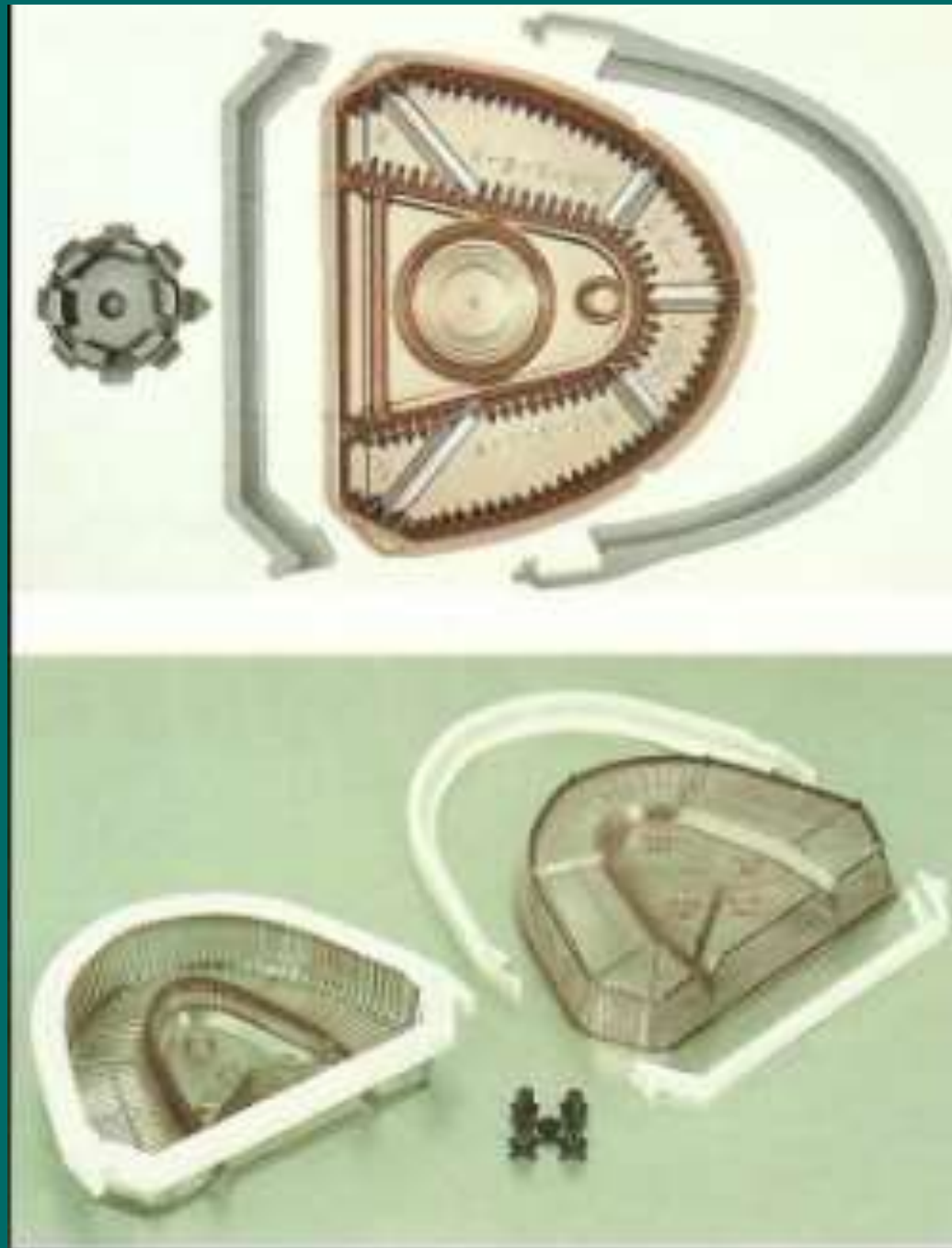


Working Cast With Removable Die

3- Di-lok tray

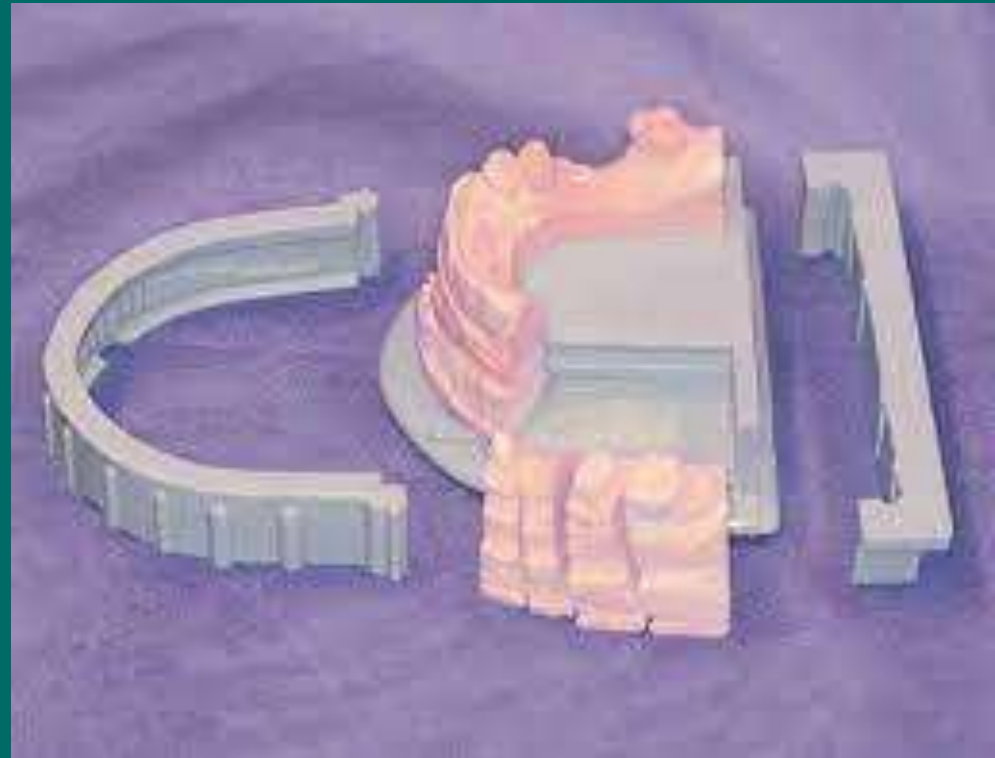
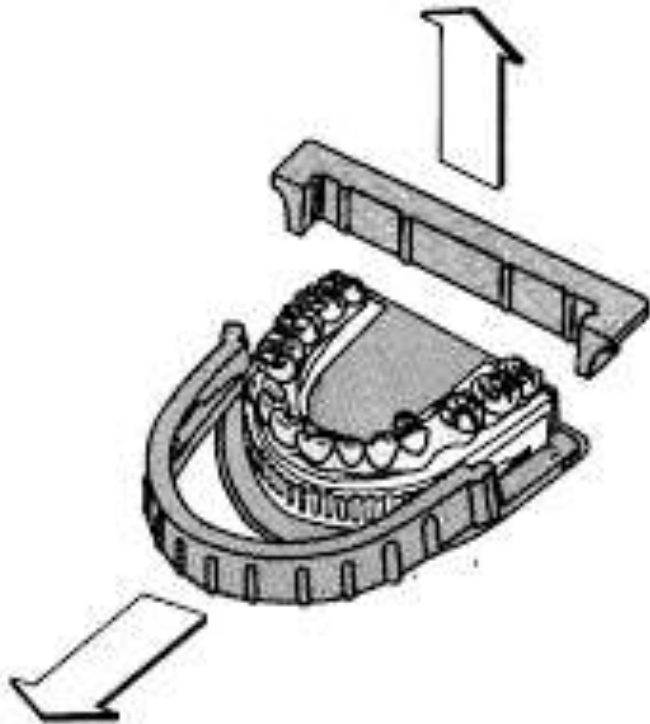
DI-LOK system:

- It is plastic tray with internal orienting grooves and notches used to reassemble working cast and dies.



DI-LOK system:

- It is made from multiple components which can be assembled or disassembled



DI-LOK system:

- When stone has set for 1 hour, separate the cast from impression.
- Cast is trimmed in horseshoe configuration to fit in Di-lok tray and the buccal border is tapered toward the base using arbor band.
- The cast should be poured in a U shape with no stone in the center, building it up to 2.5cm(1inch).
- A lingual side is trimmed with an arbor band.
- Horizontal grooves are cut in the base for retention.



DI-LOK system:

- Soak the base of the cast in water for 5 min.
- Pour the cast in tray filled with stone until $\frac{3}{4}$ of it is soaked.
- The cast seated so that the cervical line of the prepared teeth 4mm above the base level.
- Wipe off excess stone and allow it to set hard and become dry.
- Then the cast must be removed from the tray.



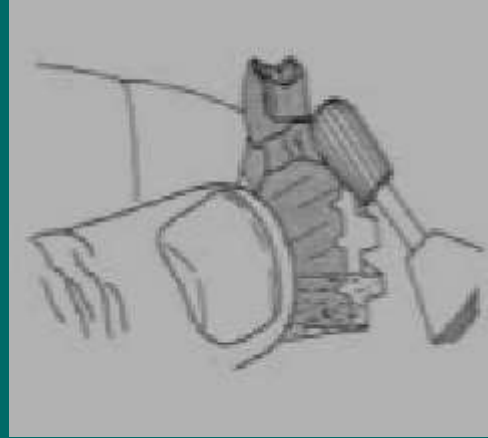
DI-LOK system:

- Disassemble the tray by lifting the back up and then slide the buccal segment forward.
- With a saw cut between prepared teeth and other teeth.
- The cut should start from interdental papilla downward with slight taper.
- The cut should extend $\frac{3}{4}$ through the stone base.
- Use finger pressure or suitable instrument to break the die from cast.

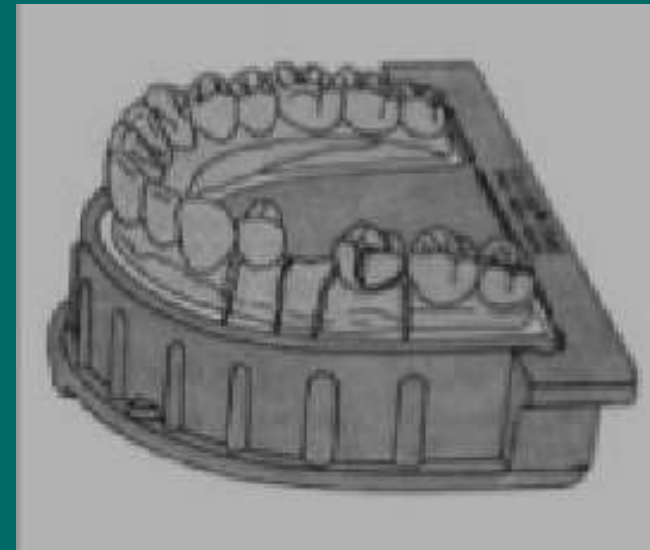


DI-LOK system:

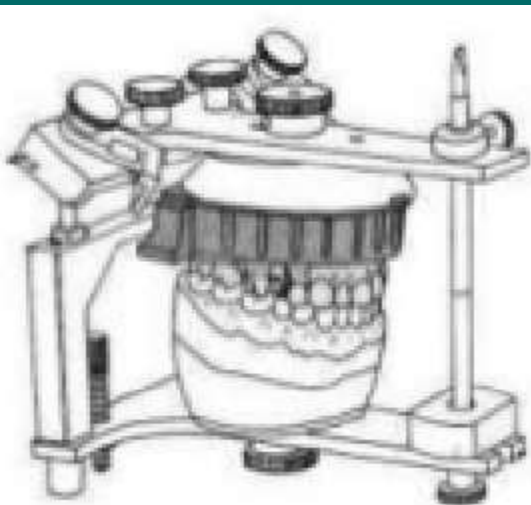
Ditching is done using pear-shaped bur.



The cast and die are reassembled in the tray.



Then the cast and tray
mounted on the articulator.



Working Cast With Removable Die

4- Pindex system



Pindex system

- In the Pindex system a reverse drill press is used to create a master cast with dies that can be removed and replaced repeatedly with great precision.



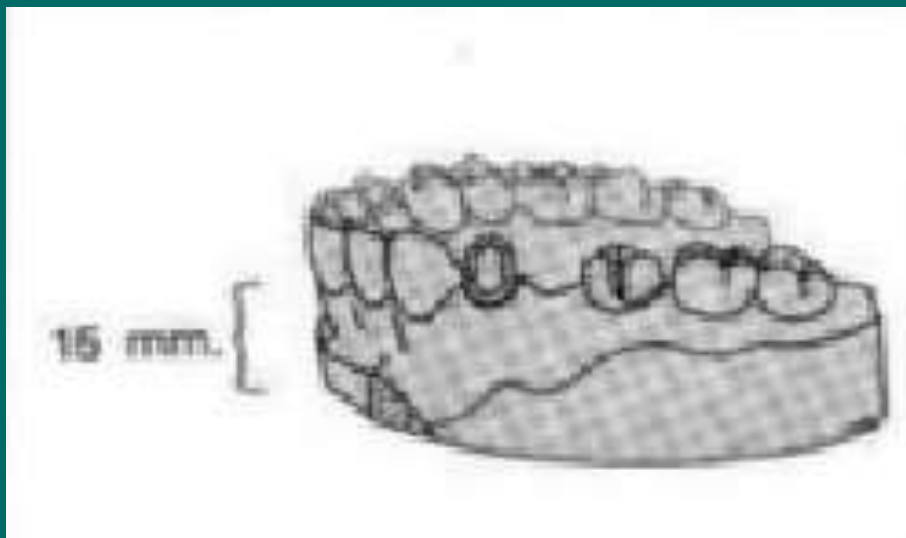
Pindex system

- The impression is poured without positioning and attaching dowel pins.
- The machine accurately drills parallel holes from the underside of trimmed cast.



Pindex system

- Pour the impression adding approximately 20mm of stone and wet the cast prior to trimming.
- Then trim the bottom of the cast resting the heels on the table of the trimmer.
- The cast should sit perfectly flat on a tabletop. Its thickness must be a minimum of 15mm.



Pindex system

- Periphery of the cast should be trimmed.
- Remove any excess stone in palate/lingual area with an arbor band on a lathe.
- The lingual border of the cast should taper slightly toward the base to facilitate removal of the dies from the cast later.
- The faciolingual width 20mm.



Pindex system

- use a pencil to mark the desired location of pins on the occlusal surface of the teeth or preparations.



Pindex system

- Place the cast on worktable and align marked teeth with the illuminated dot from the light beam director.
- Using both hands, exert firm downwards pressure on the cast with thumbs.
- Press the handle bar with slow, even pressure and the drill assembly moves upwards cutting the pin holes.

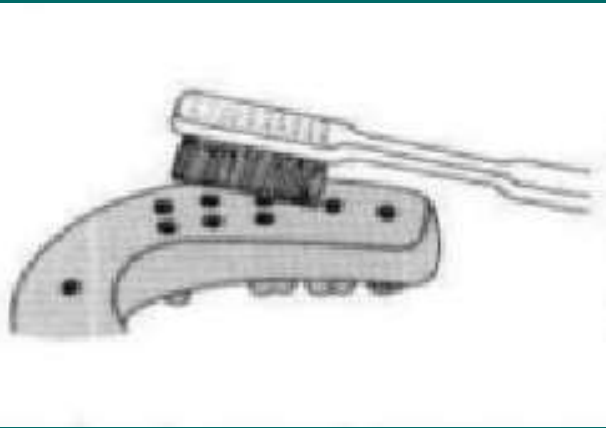


Pindex system

- When proper depth is achieved red light goes off. Same procedure is repeated with each mark.
- For better results, cast is made slightly damp to prevent dust formation.



Pindex system



Compress air and brush is used to remove debris from holes.



The pin holes are refined with hand reamer.

Cyanoacrylate cement is placed on the pins prior to cementing the pin tips.

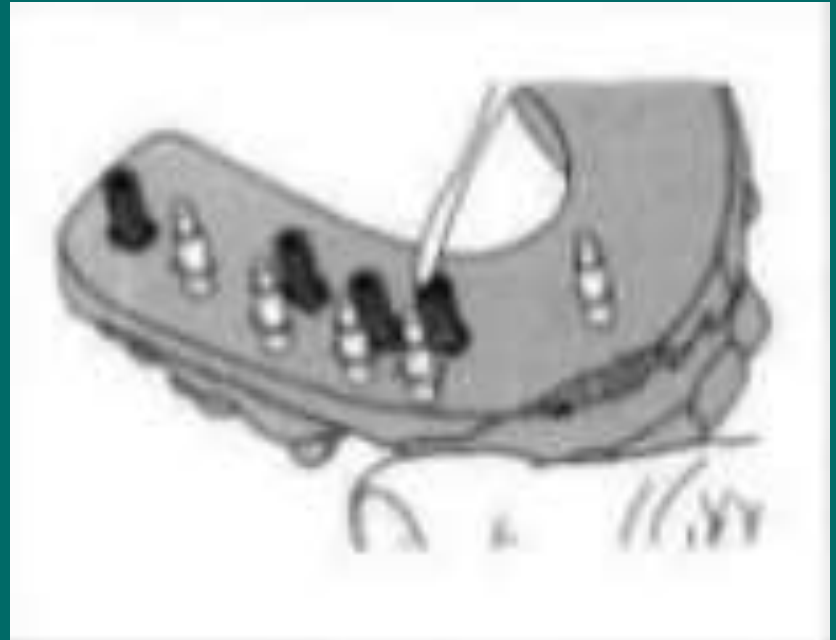


Shorter pins are placed before the long pins
In lingual or palatal holes and long pins in the
facial holes.



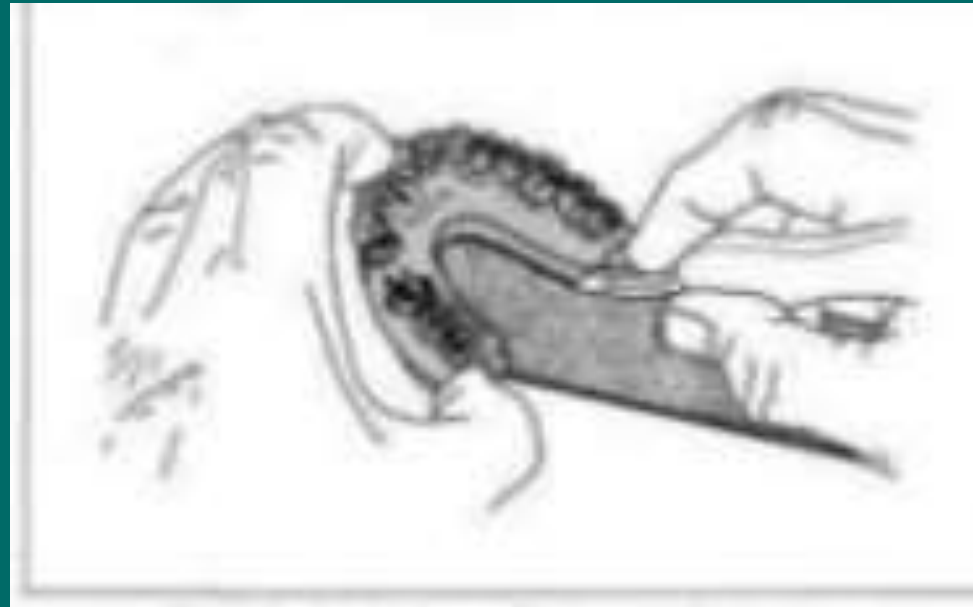
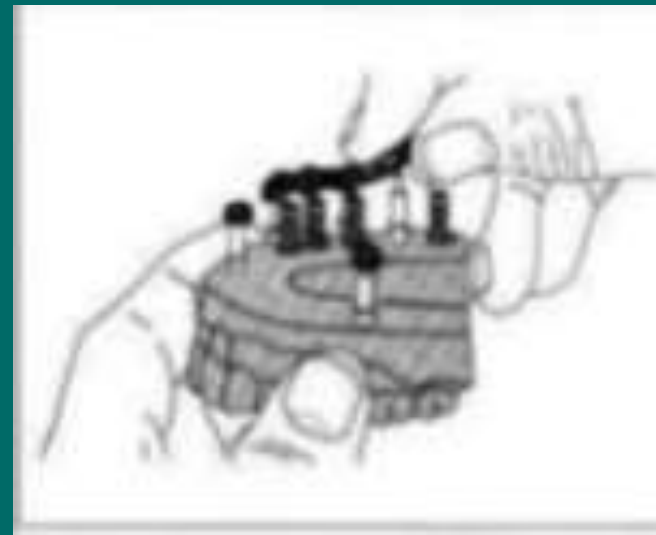
Pindex system

- Sleeves with certain color (white) are placed on long pins and sleeves with different color (gray) are placed on short pins.
- The bottom of the cast is slightly coated with separating medium.
- Gray sleeves are blocked with molten wax to prevent the sleeve from filling with stone when base is added.



Pindex system

- Strip of utility wax along the ends of the long pins is added to facilitate removal of the dies later.
- A palatal or lingual filler of boxing wax is fixed in place.
- The cast is seated over a mould filled with stone.



Pindex system

- When stone becomes hard, saw cuts are premarked with the pencil.



Then dies are sectioned from the underside after separating the cast from the base.

It may also be sectioned from the occlusal aspect.



Pindex system

- After the dies are sectioned, trim them in a conventional manner.
- Mark the finish line with red pencil.
- Apply die hardener and die spacer.
- Then place the completed cast on the articulator and the cast is ready for fabrication of the wax pattern.

Working cast with separate die

1- multi-pour technique

Working cast with separate die

ADVANTAGES:

- Simple.
- Slightly more accurate.
- Minimum trimming.

Disadvantages:

- Difficult to transfer complex or fragile wax pattern from cast to die.
- Seating the pattern on the cast may be problematic.
- Technique can be used with elastomeric impression materials only.

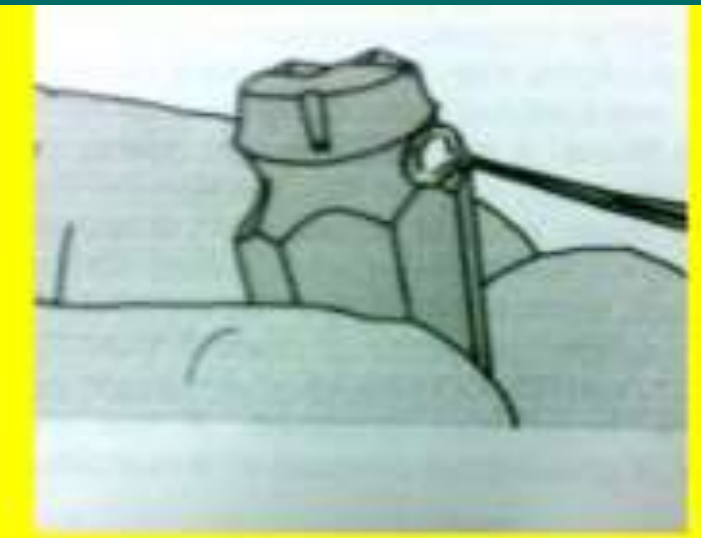
multi-pour technique:

- The area of preparation is poured with stones in small increments
- When stone set hard, it is separated and second pour is then made of the entire impression.
- The first pour which is more accurate is trimmed into a die with handle.
- A properly trimmed die slightly larger in diameter than the preparation.



multi-pour technique:

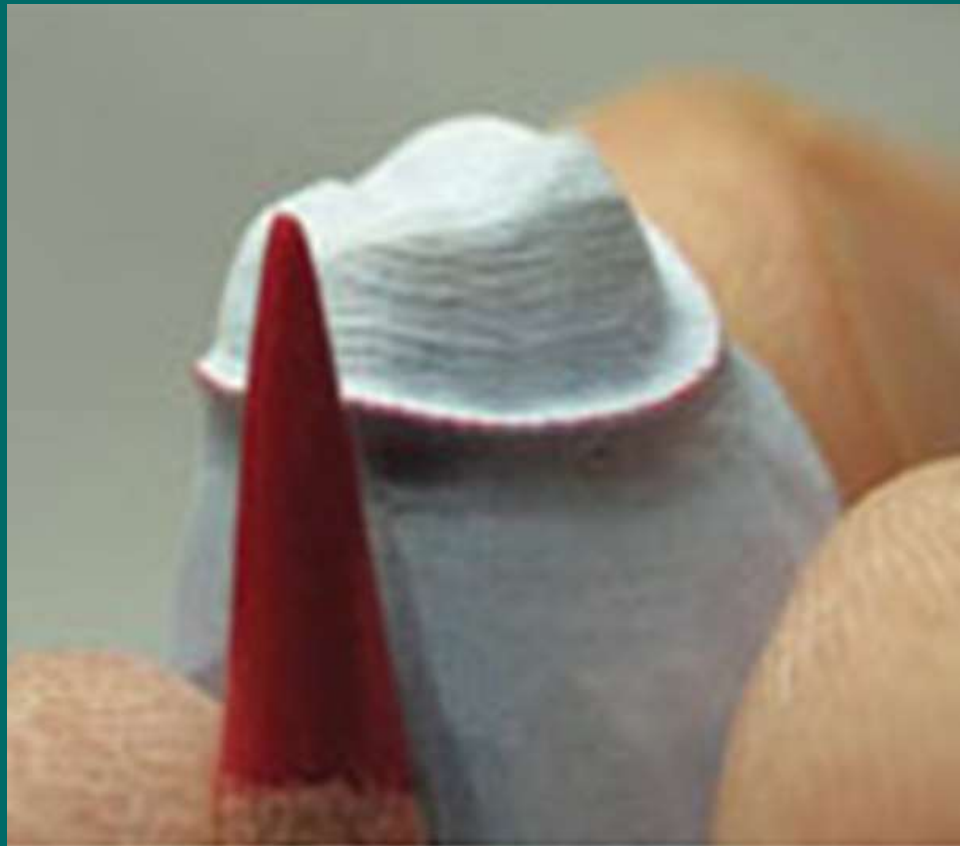
- Improperly trimmed die with a handle that meets the preparation at an angle.
- Handle should be 1 inch long.
- The die is trimmed with an acrylic bur.
- Shaping of the handle near finish line is completed with a scalpel.



The die is smoothed below finish line with the discoid end of a tanner carver.

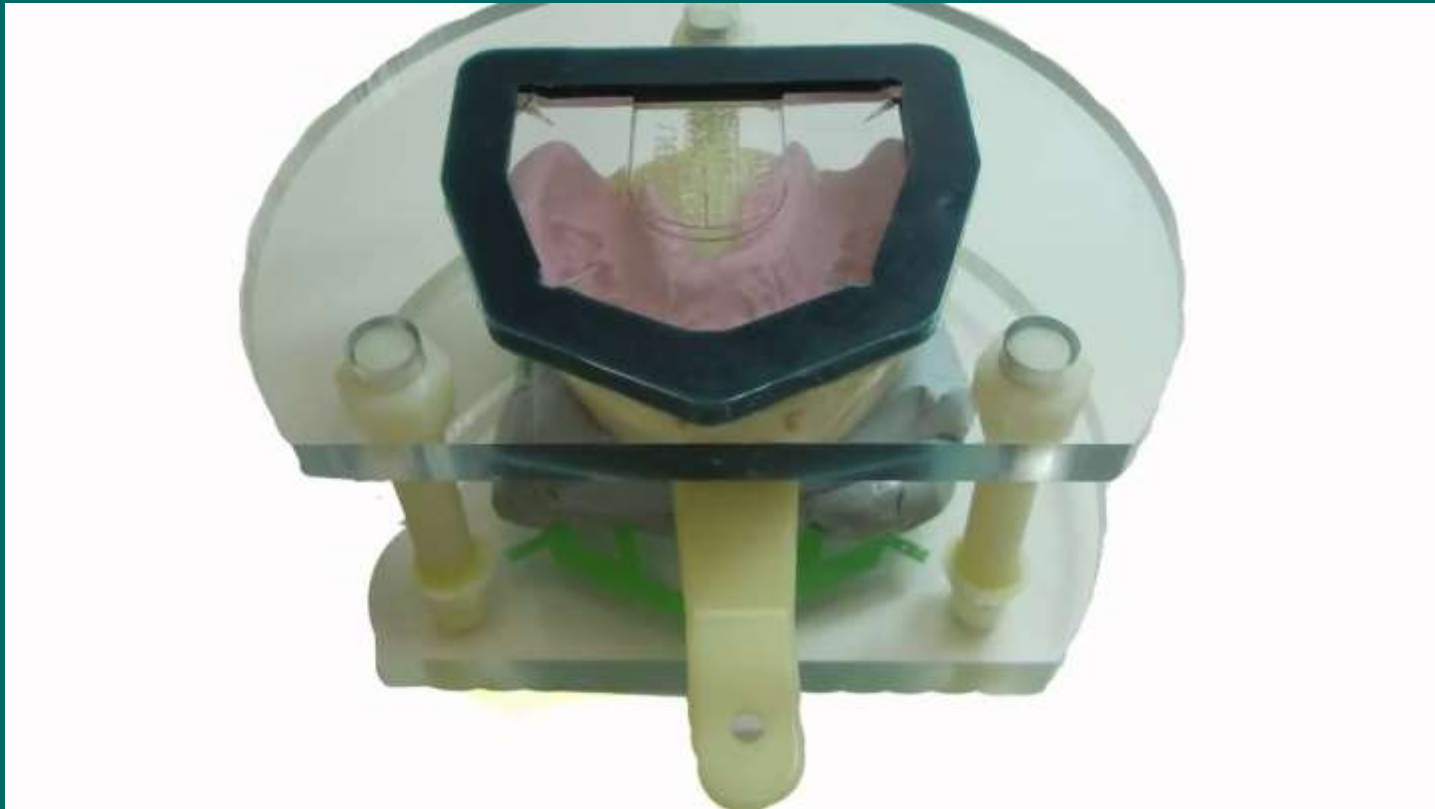
multi-pour technique:

- The prepared finish line on the die should be outlined with a red pencil.
- Die relief agent is painted on the preparation.

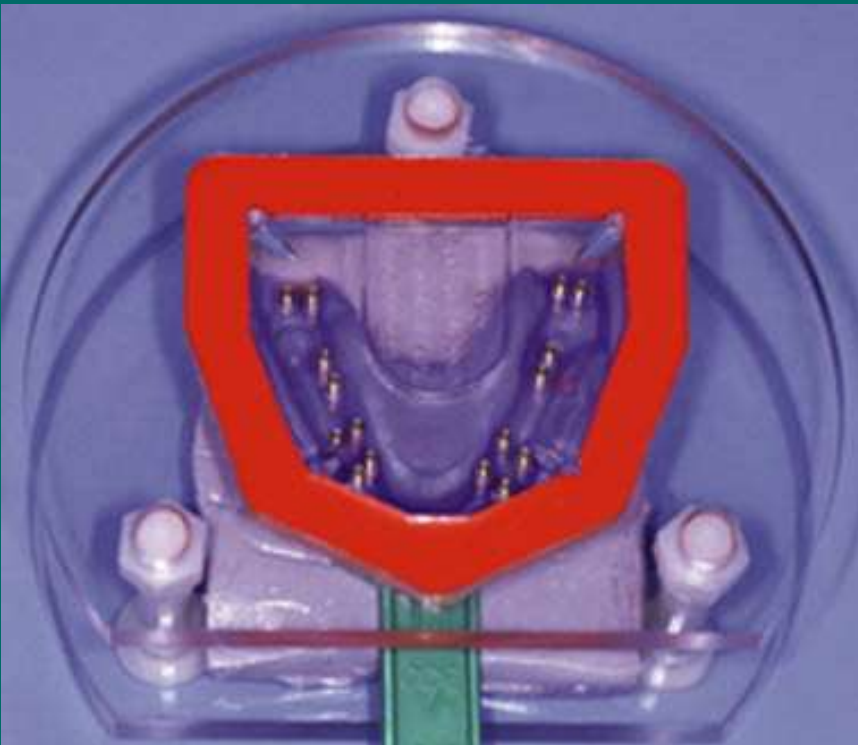


**Working cast with
separate die**

DVA model system



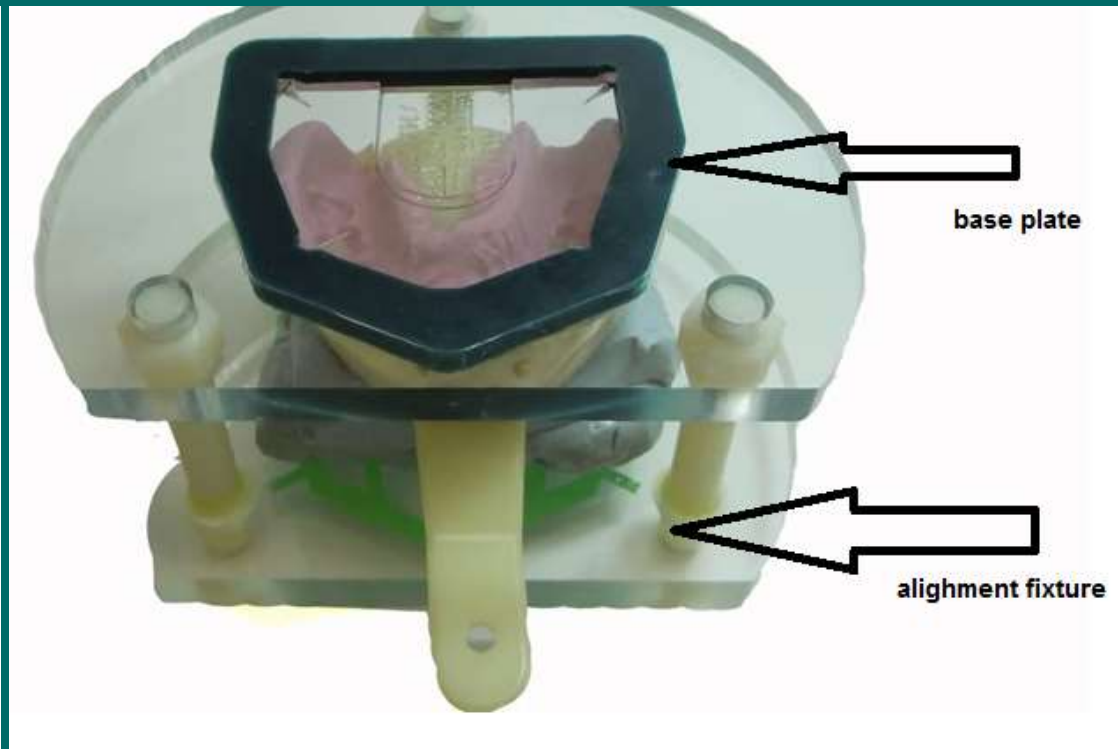
2- DVA model system



DVA model system:

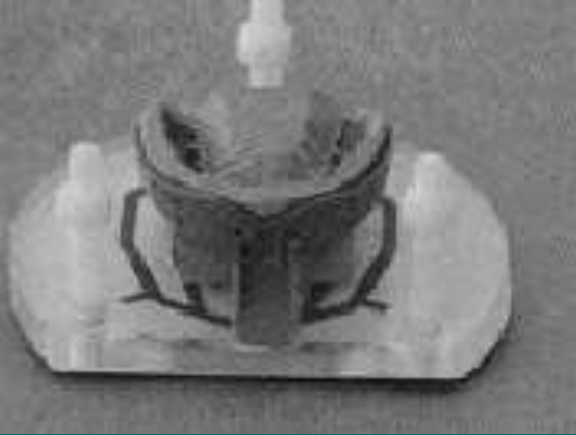
- DVA model system is composed from :

1- alignment fixture. 2- base plate. 3- drilling machine.



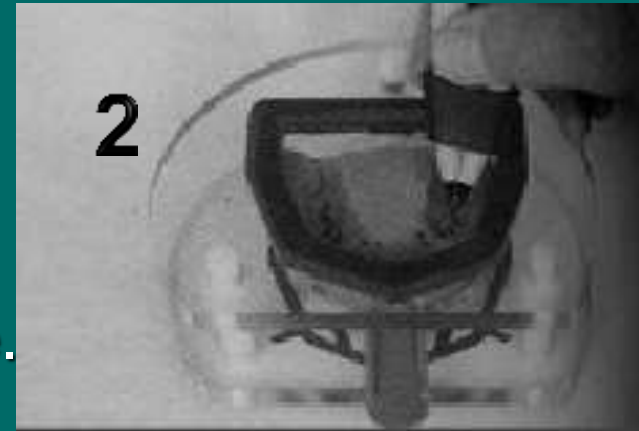
DVA model system

1



1- Trimmed impression on alignment fixture.

2



2- Marking dowel pin locations on clear plate.

3- Drilling holes for dowel pins as marked.

4- Inserting dowels in the baseplate.

4



3



2- DVA model system

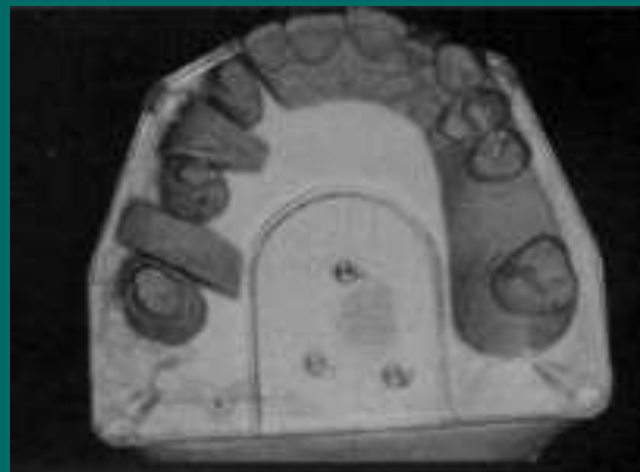
- Impression is poured, stone placed around dowel pins and alignment fixture (base plate) replaced over poured impression.
- Set cast is removed from base plate



2- DVA model system



Cast is trimmed



Cast is sectioned



Dies are trimmed and working casts mounted on articulators.

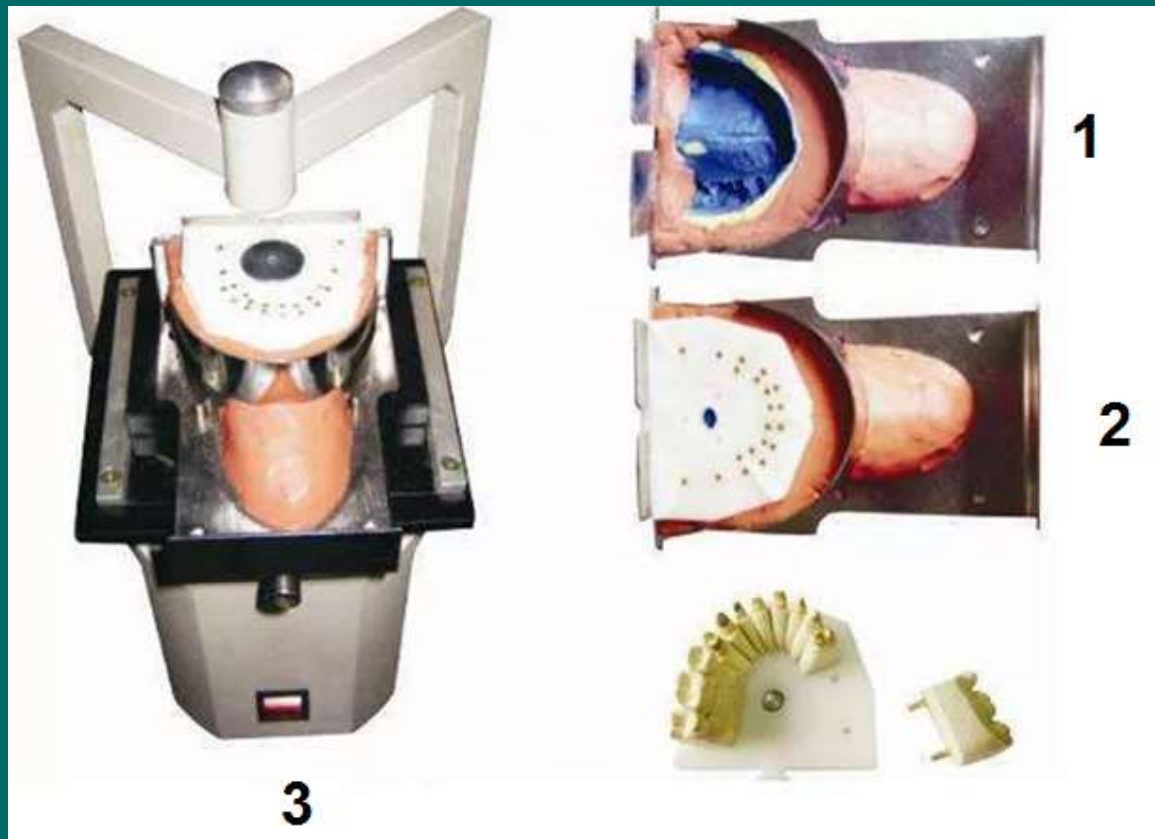
Working cast with separate die

3-Zeiser model system

3-Zeiser model system

- Zeiser model system is composed from:

1- Impression alignment. 2- pin alignment (fixation) plate. 3- pin holes punching machine.



3-Zeiser model system

- Impression is leveled, blocked out with silicone putty and positioned over baseplate.
- Pin locations are determined and the pin holes drilled in the base.



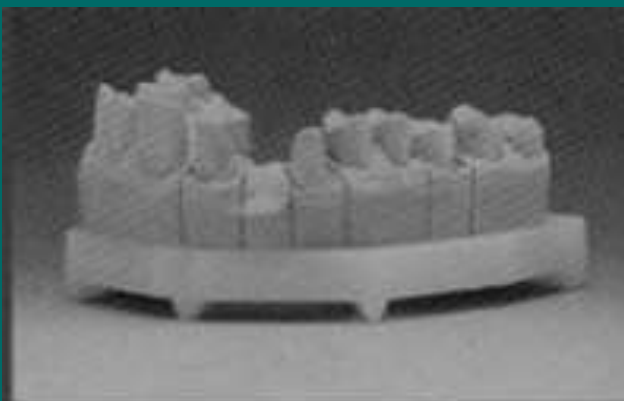
3-Zeiser model system

- Pins are inserted into the base.
- Impression is poured.
- Base is inverted into the stone.



3-Zeiser model system

- Cast is separated from the impression when set and then separated from the base.
- Precision saw aids sectioning.



Sectioned cast



New techniques for producing working casts:

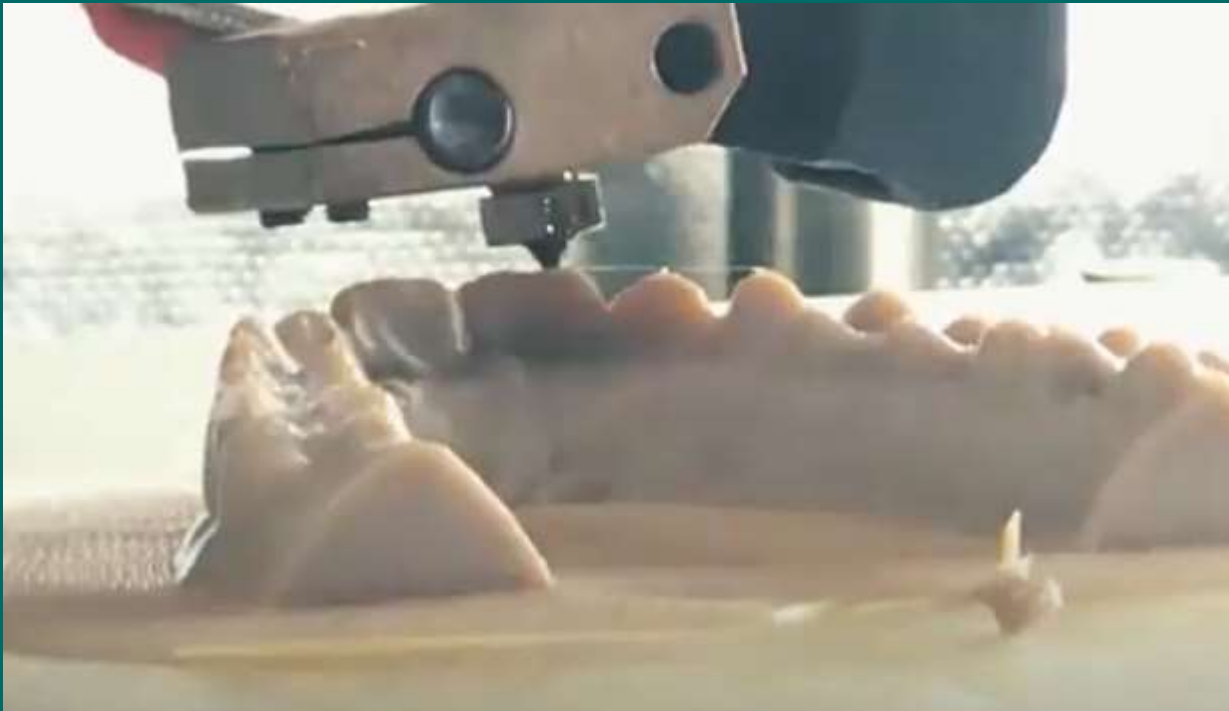


3D printing for dental models



3D printing for dental models

- After loading the 3D printing machine with the scan of impression or intraoral camera information, the machine will build the model automatically in highly perfect way.



3D printing for dental models

- The model is seated in oven for curing



3D printing for dental models

- After hardening and cooling of material the model is removed and trimmed.



3D printing materials

Flexible, TPE, TPU Thermoplastic Urethane / Polyurethane						
FPE Flexible Polyester						
Glow-In-The-Dark Glow in the dark PLA or ABS						
HIPS High Impact Polystyrene						
Lignin (bioFila) Lignin and PLA plus additives						
Magnetic PLA with powdered iron						
Metal PLA / ABS Metal Powder and PLA or ABS blend						
nGen Similar to PETG						

BIBLIOGRAPHY

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2. Rosensteel (2001) Contemporary Fixed Prosthodontics 3rd edition Mosby
3. Shillenbergl (1981) Fundamentals of fixed prosthodontics 3rd edition Quintessence
4. Silver plated dies marginal accuracy of cast restorations J Prosth Dent 51;1984: 768-772
5. Comparision of surface detail reproduction of flexible die materials J Prosth Dent 1998; 80:485-9
6. An investigation of epoxy resin dies J Prosth Dent 1980;44:45-9

Types of Wax

- Type I inlay casting wax.
- Type II inlay casting wax; it is a softer wax than the type I and is used for extraoral wax technique. It has a lower melting temperature. It is used to construct the wax pattern on the cast.

Techniques of construction of wax pattern

1- Intraoral Technique (Direct Technique).

The wax pattern is constructed inside the patient mouth using type I inlay wax. This technique is mostly used to construct the posterior inlay restoration and anterior post-crown.

2 - Extraoral Technique (Indirect Technique):

Type II inlay wax is used to construct the wax pattern on the die of the working cast.

Steps in waxing procedure

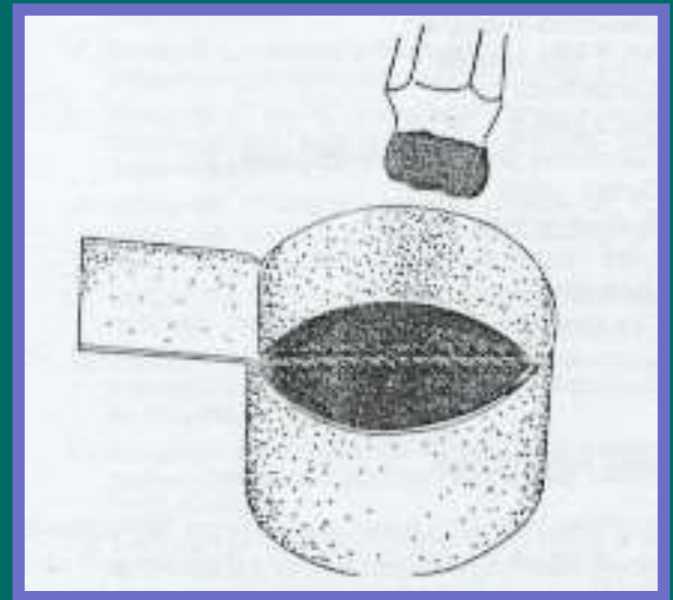
- Apply die spacer on the die within 1mm from the finish line and leave it to dry (obtain a relive of 20 to 40 microns).
- Then apply die lubricant (separating medium) to the die.



Steps in waxing procedure

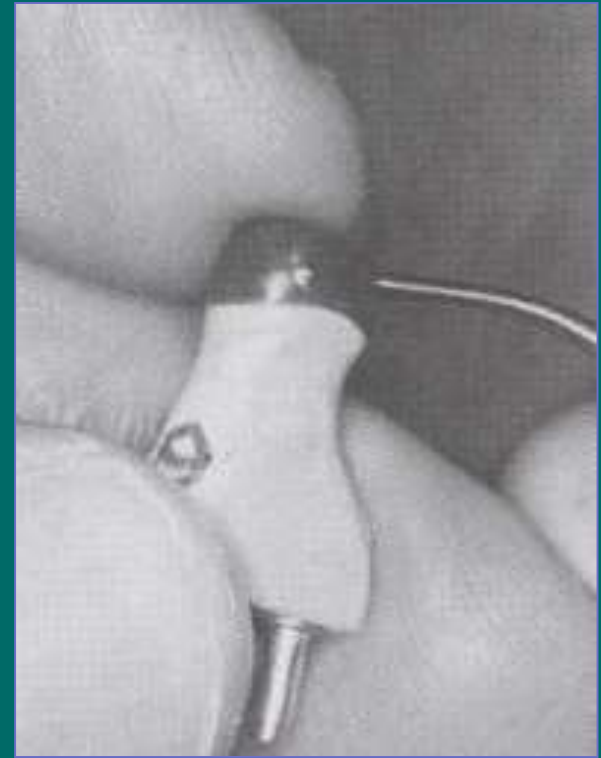
Cover all the preparation in one time with molten wax to make a coping of wax.

This can be obtained by immersing the preparation in a dish containing molten wax (this is the first layer of wax pattern).



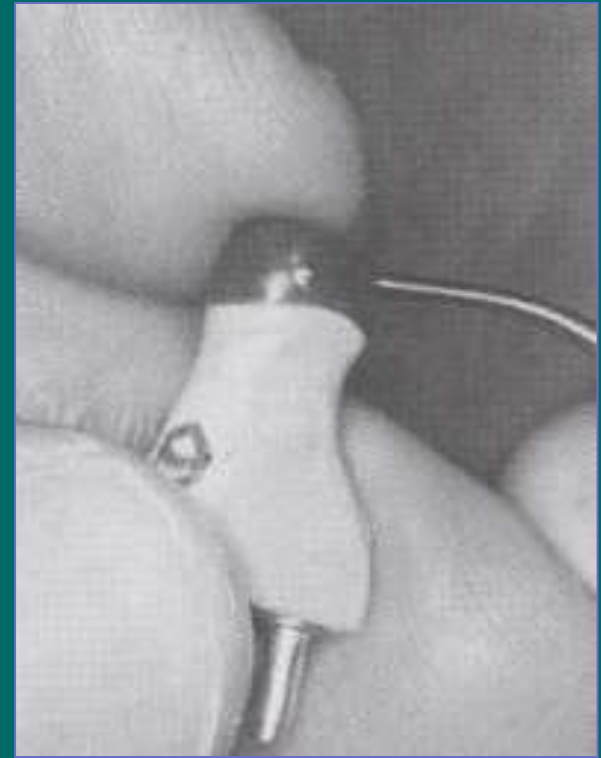
Steps in waxing procedure

- Add wax to the proximal surfaces of the preparation to build the proximal surfaces and contact areas with the adjacent teeth.



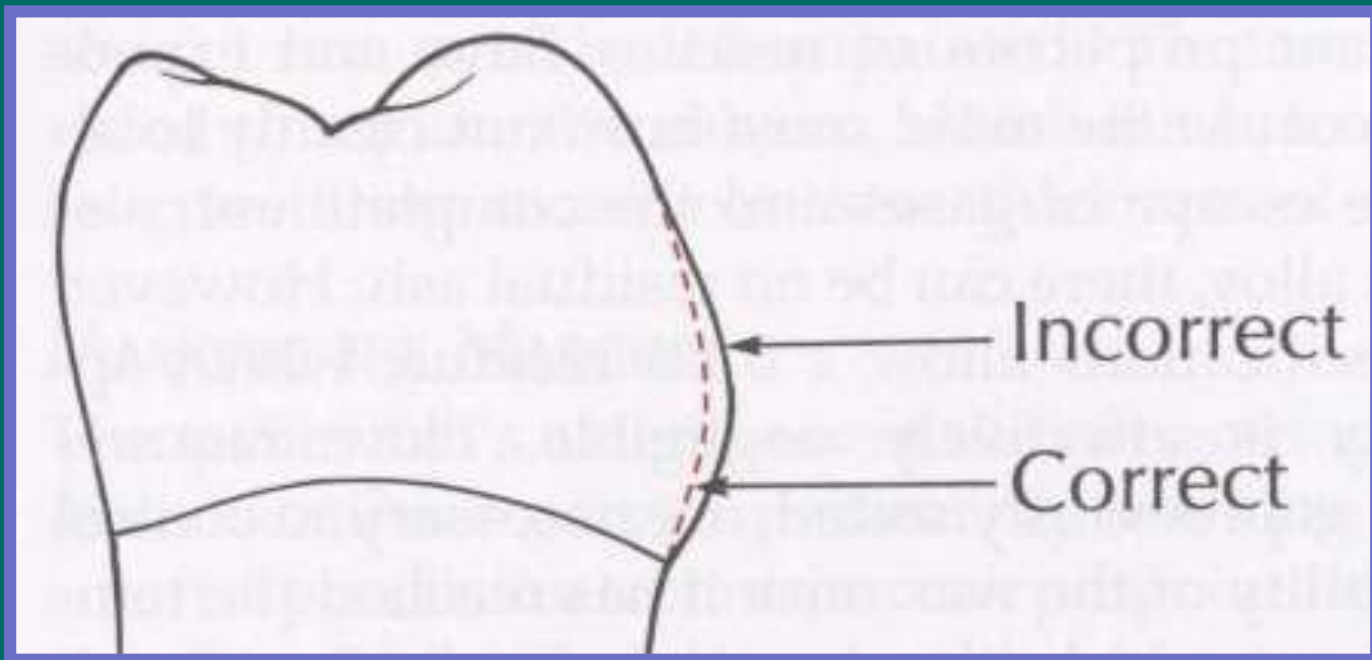
Steps in waxing procedure

- Add wax to the proximal surfaces of the preparation to build the proximal surfaces and contact areas with the adjacent teeth.



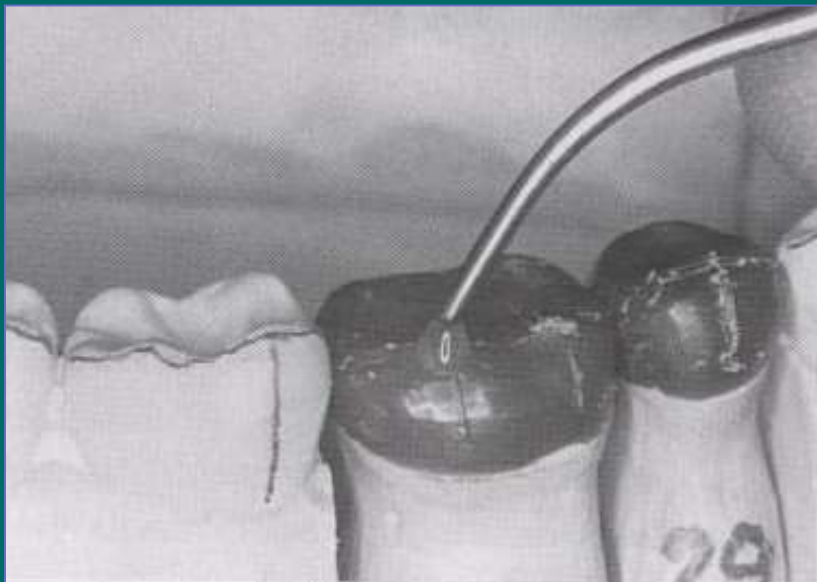
Steps in waxing procedure

- Build the axial wall (buccal and lingual wall) to the normal contour.



Steps in waxing procedure

- Build the occlusal surface of the restoration following the curve of spee and curve of Wilson, then check and adjust the occlusal relation with the opposing teeth in centric and eccentric relation



Steps in waxing procedure

- Check the margins of the wax pattern to ensure that the margins of the wax pattern have no over or under extension.



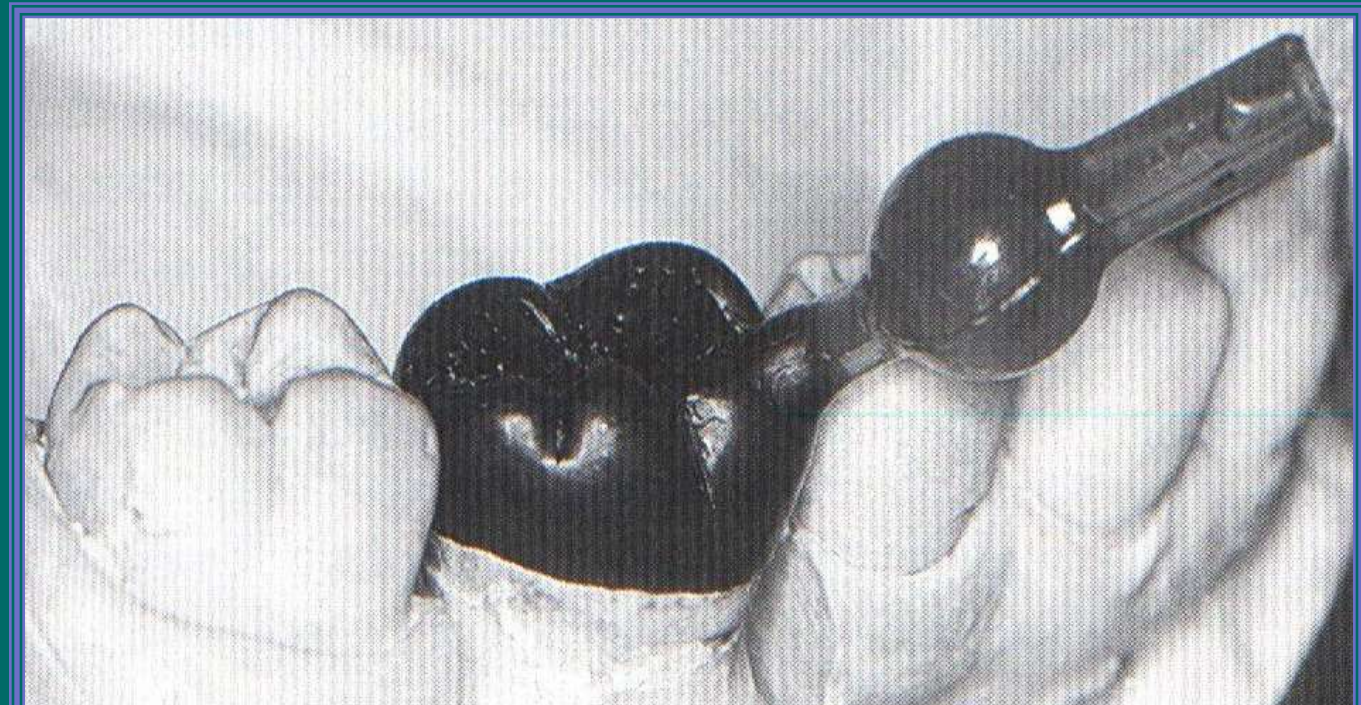
Steps in waxing procedure

- Check the margins of the wax pattern to ensure that the margins of the wax pattern have no over or under extension.



SPRUE

- ✓ It is a small diameter pin made of wax, plastic or metal.
- ✓ one end of it is attached to the wax pattern while the other end is attached to the crucible former.
- ✓ It provides a channel after burn out procedure to act as inlet for the gold which is forced in the mold cavity.



thank
YOU
SO MUCH
😊